DEUTSCHE BUNDESBANK
EUROSYSTEM

# Introduction to Machine Learning

Deutsche Bundesbank – March 2026

International Central Banking Courses

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# Structure and Organization of the Course

# Structure of the Course

## Online Sessions

Day 1: March, 2

- Machine Learning Concepts (Quick Recap)
- Interpretable and Explainable Machine Learning
- Causal Inference

Day 2: March 3

- Anomaly Detection
- Natural Language Processing
- Orientation for the Use Case Development

# Structure of the Course

## On-Site Sessions

Day 1: March, 17

- Introduction
- Overview of Use-Cases (Use-Case Pitches, Guided Discussion in Breakout Groups)
- Use-Case Sprint I

Day 2: March, 18

- Use-Case Sprint II
- Feedback Coffee
- Use-Case Sprint III

Day 3: March, 19

- Open Coaching Session
- Short Presentations by Participants
- Wrap-Up

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# What are your expectations regarding this course?

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# How would you rank your knowledge in...

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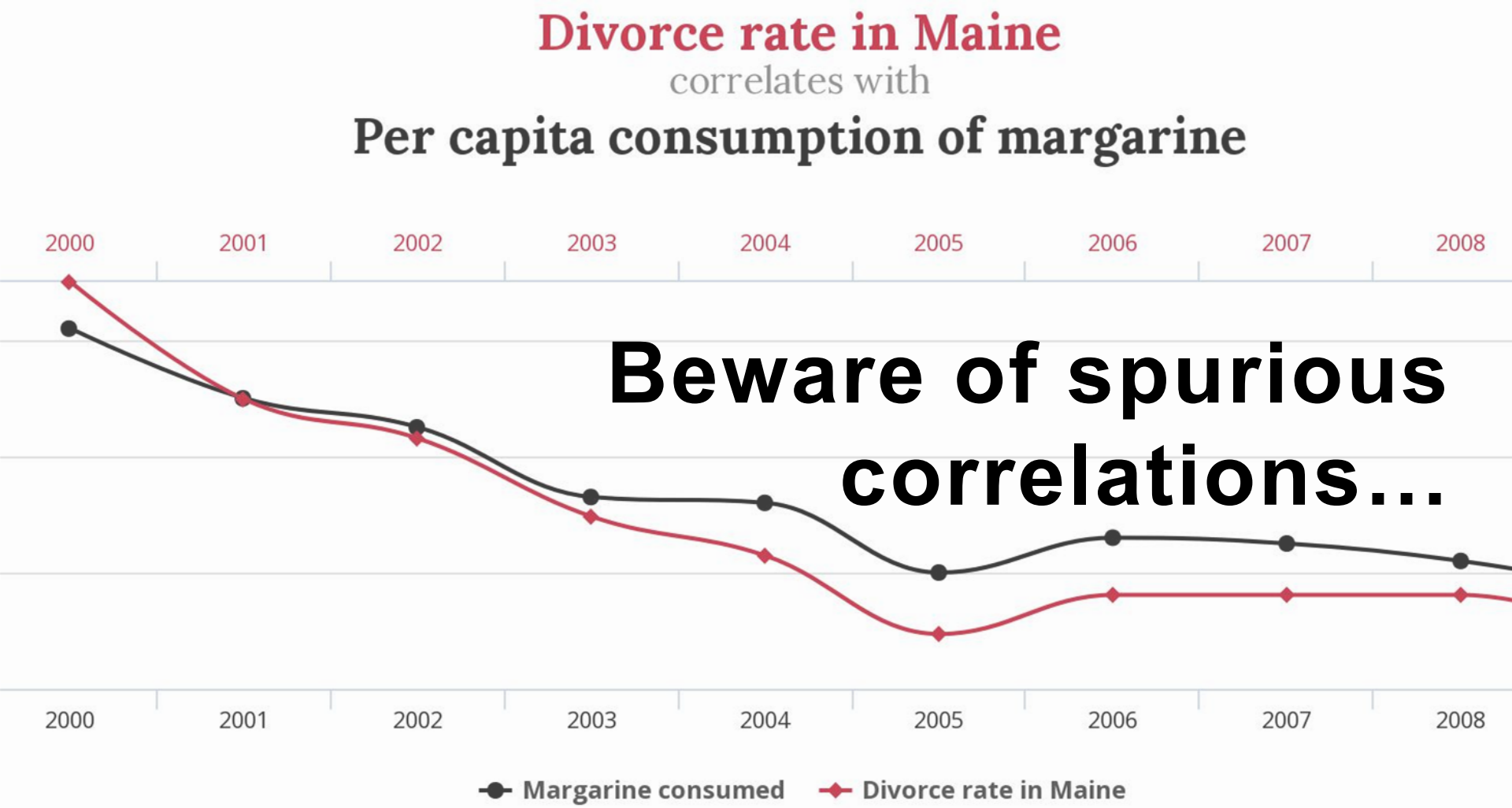
# What challenges can we face when training machine learning models?

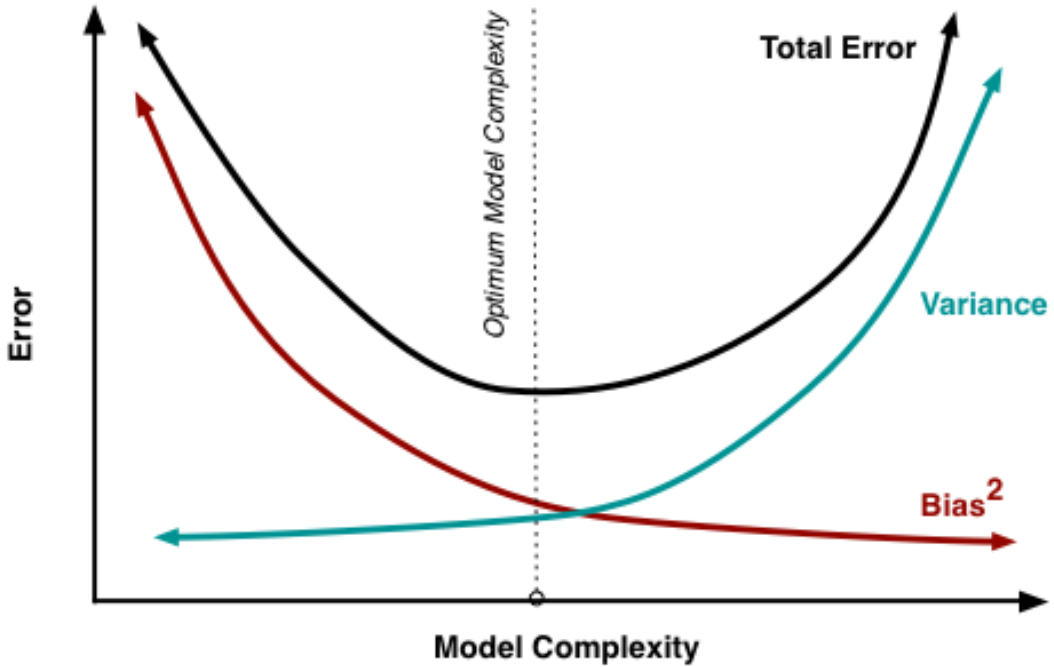
# Machine Learning Concepts (Quick Recap)


Divorce rate in Maine
correlates with
Per capita consumption of margarine
2000
2001
2002
2003
2004
2005
2006
2007
2008
Beware of spurious correlations…
Margarine consumed
Divorce rate in Maine

# Train, Validation, and Test Subsamples
# Splitting the Data

- Golden rule: evaluate the model on **data** that **was not used** to **train** it
- Most common: **randomly** split the dataset into **train and test** data
  - Usually: **75%-80%** train, **25%-20%** test
- Better approach: **randomly** split dataset into **train**, **validation**, and **test** data
  - Usually: **60%** train, **20%** validation, **20%** test
- Even better in some settings: also have **out-of-time** data
  - **Completely separate** dataset, e.g., collected a few months later
- **Cross Validation** is even better, if feasible

# Bias-Variance Trade-Off

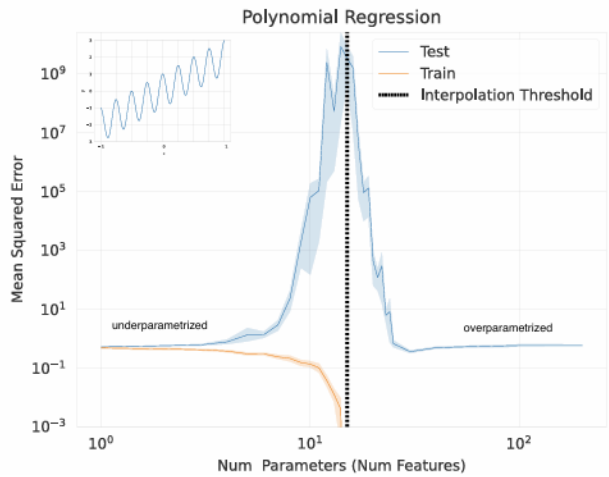



Source: scott.fortmann-roe.com/docs/BiasVariance.html

**Any exceptions?**

# Double Descent

- Test error keeps dropping with an overparameterized model (e.g., Large Language Models)

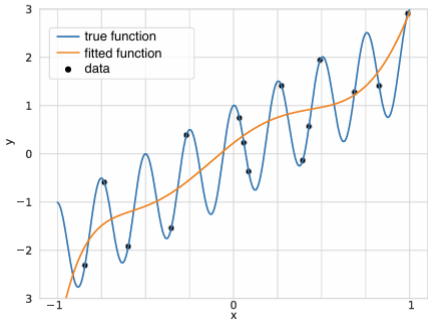



Source: Schaeffer et. al (2023)

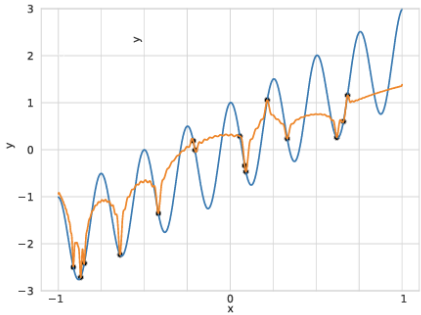
- How/when does this happen? Set of special conditions depending on:
  - Number of data points
  - Dimensionality of the data
  - Number of model parameters

# Double Descent

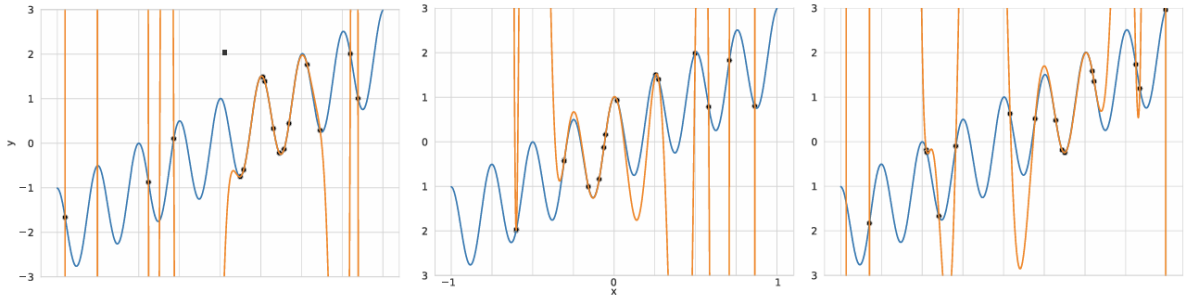
underparametrized




overparametrized




interpolation threshold

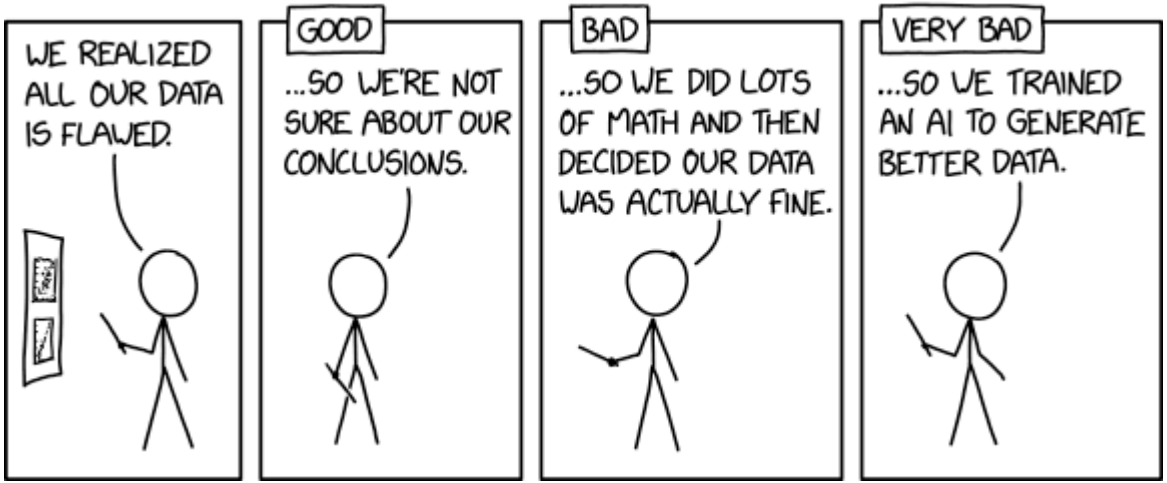



Source: Schaeffer et. al (2023)

# Time Invested in Data Preparation is Time Well Spent

- Regardless of how “advanced” the method you are using is: “garbage-in, garbage-out” applies

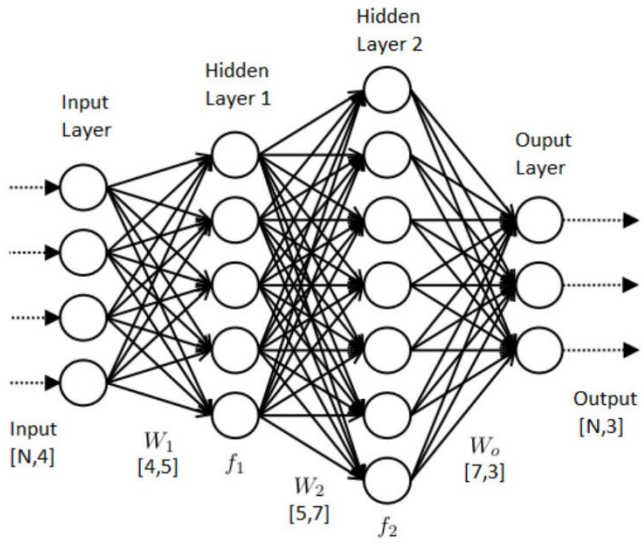
- In times of artificially generated data (e.g., texts) – having access to “bio” data is crucial




Source: https://xkcd.com/2494/

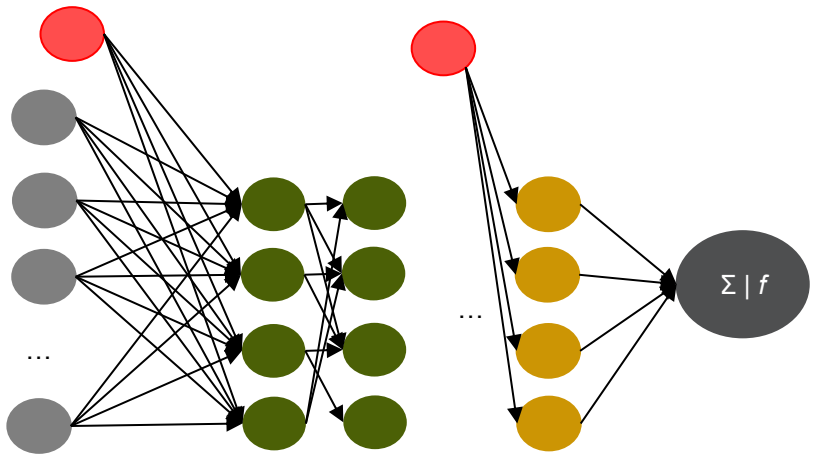
# Neural Networks

- Every node is a **weighted sum** of previous nodes
- Extra **layers** can add **complexity or structure**
- Weights are **iteratively improved** using gradient descent
- Output layer **converts** the **high-dimensional representations** into the final output:
  - Classification probabilities
  - Regression results
  - Sequences for language generation
  - …



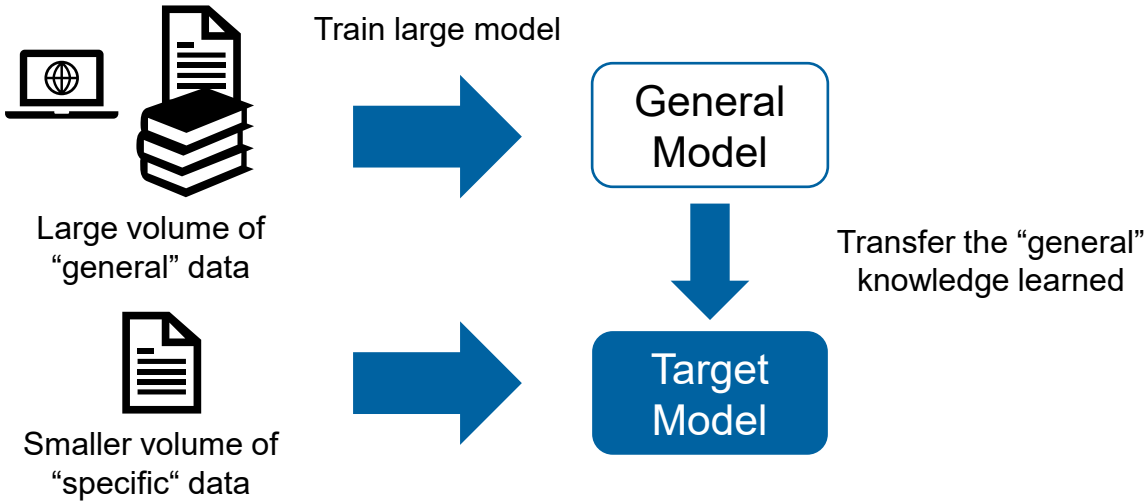
# Practical Considerations with Neural Networks

- Data **transformation** (e.g., scaling): avoids distorted weights and helps conversion
- **Synthetic** data generation and data **augmentation**: helps to learn from similar cases
- **Regularization**: helps against overfitting



# Leverage Transfer Learning

- Transfer learning: build upon existing (large) models that are expensive to train
- Possible to fine-tune them: customize “general” models to apply them to your specific use-cases



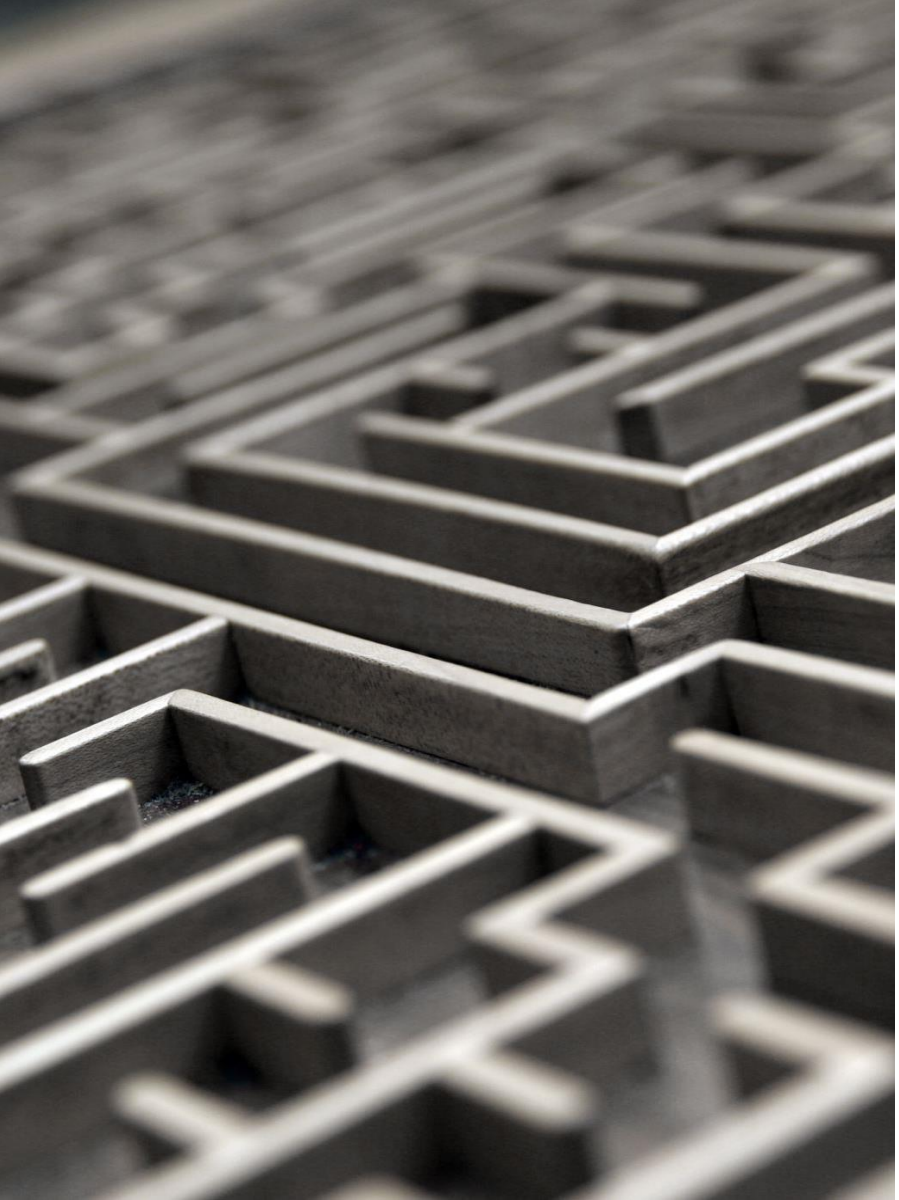
# Transfer Learning with Transformers

- Neural networks with **attention mechanism** (Vaswani et al., 2017)
- **Track relationships** in sequential data (e.g., words in a sentence), also if they are far apart (short and long-distance relationships)

- Find patterns – **no need for labelled data**
- General-purpose model **pre-trained on a large volume of data**
- **Richer and more accurate** representation of language **contextual relationships**

- **Fine-tune** with domain-specific data for several tasks (e.g., information retrieval, sentiment analysis, answering questions, …)

# Interpretable / Explainable Machine Learning



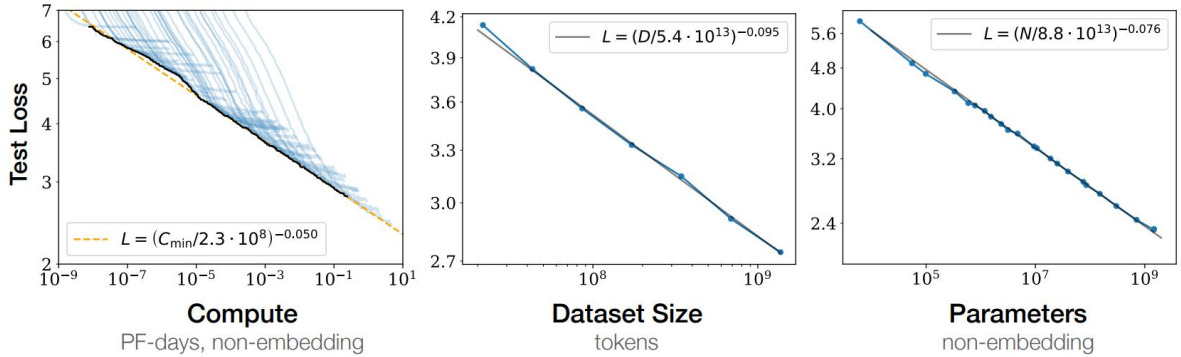
*“Nobody understands how modern AI systems do what they do.*

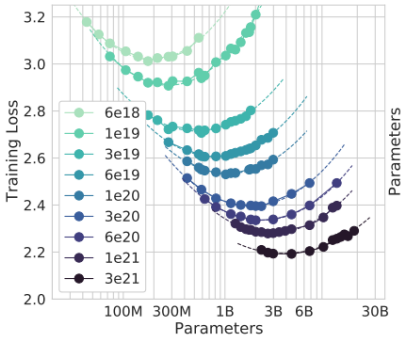
*They are giant, inscrutable matrices of floating-point numbers that we nudge in the direction of better performance until they inexplicably start working.”*

Eliezer Yudkowsky

# Models Keep Getting Larger

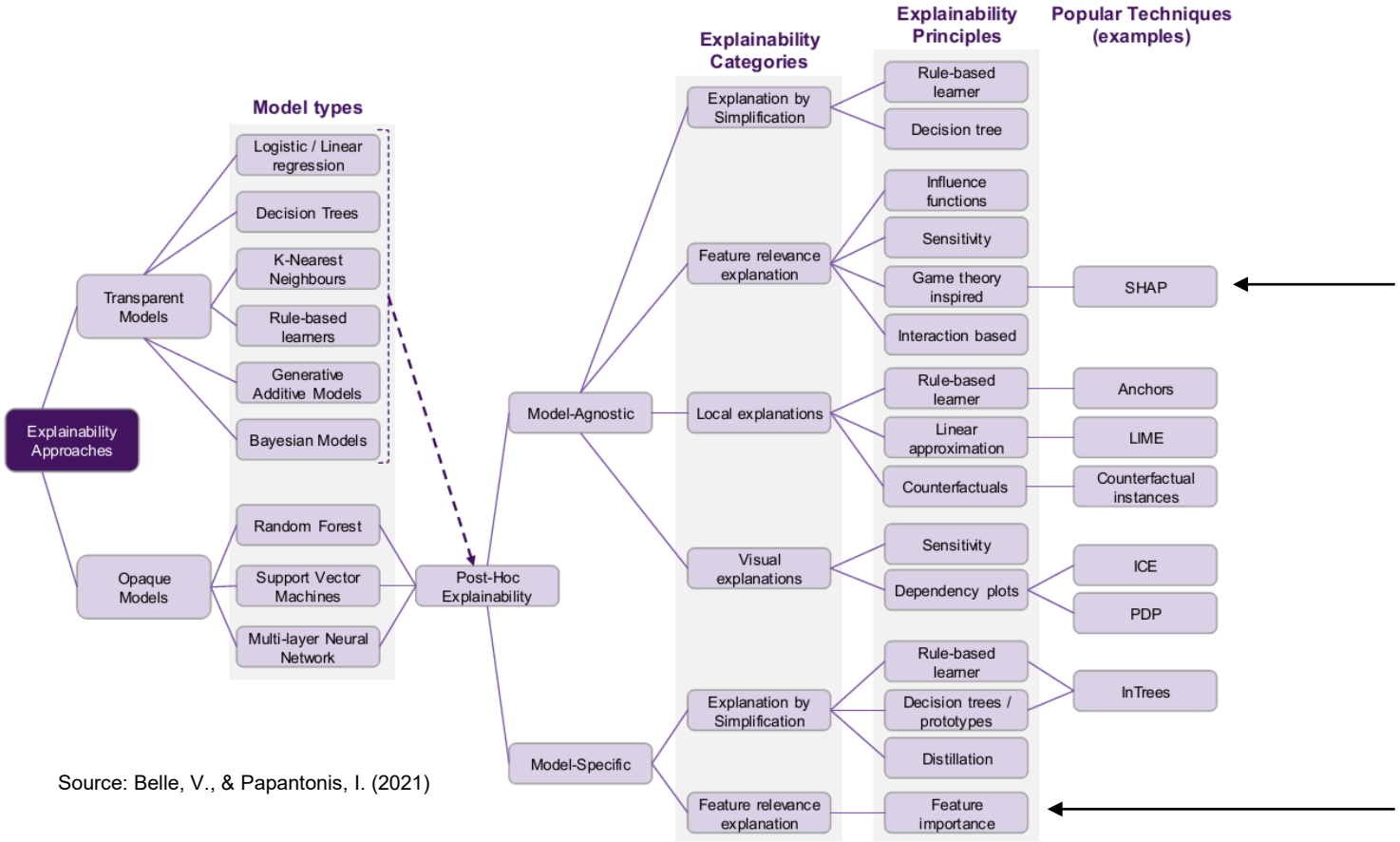
- Current machine learning models are huge (tens of billions of parameters)
- Empirical scaling laws imply increasing computing power, data, and model size will lead to even better performance




Sources: Kaplan et al. (2020), Hoffmann et al. (2022)

# Taxonomy of Explainability Approaches



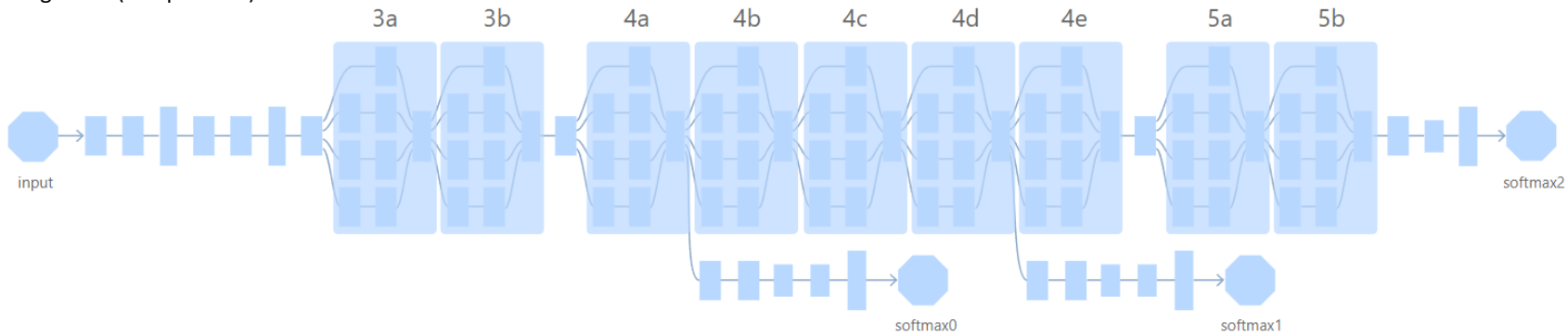

Source: Belle, V., & Papantonis, I. (2021)

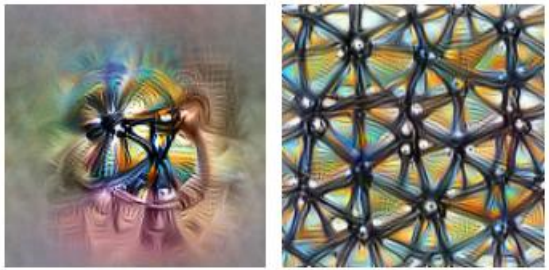
# Feature Relevance Explanation: Mechanistic Interpretability

- Even if a model makes successful predictions, it is unclear how it arrives there
- “Hidden” neurons are difficult to understand. Possible ideas:
  - Investigate at extreme-value inputs (how do they change the result?)
  - Optimizing an input for maximum activation (while maintaining transformation robustness)
- Mechanistic interpretability research
  - “[…] *reverse engineering the computational mechanisms and representations learned […]*” (Bereska, L., & Gavves, E. 2024)
  - Similar to natural science: investigating, making predictions and testing them experimentally

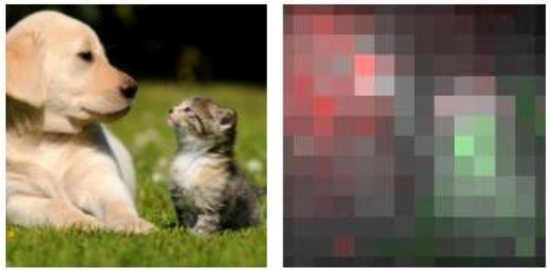
# Feature Relevance Explanation: Mechanistic Interpretability

GoogLeNet (Inception v1)



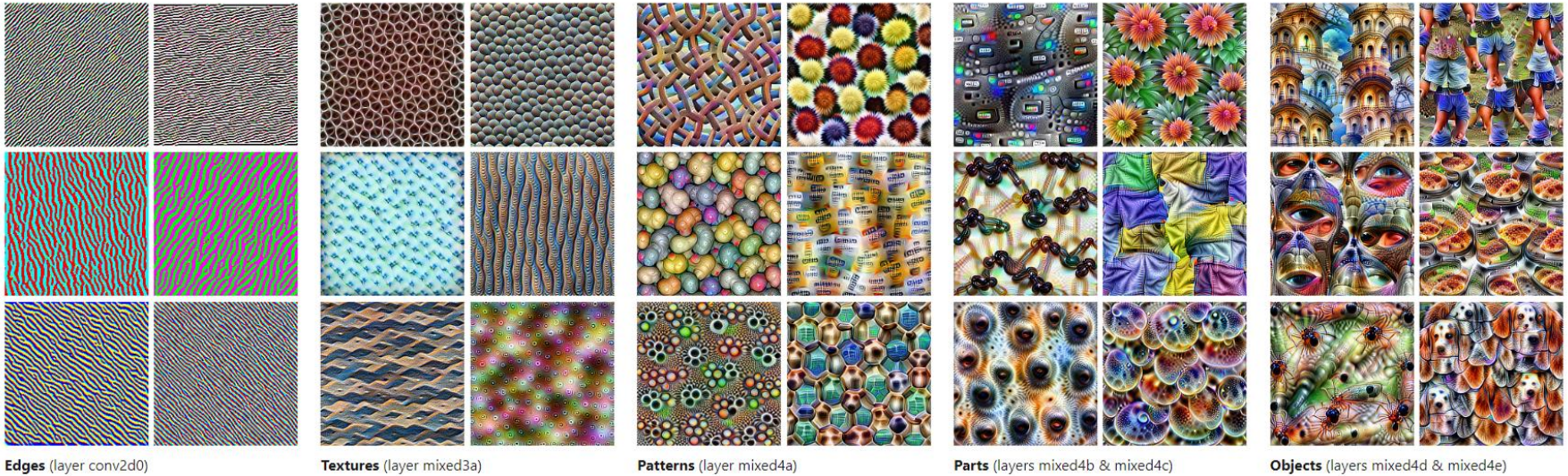
**Feature visualization** answers questions about what a network — or parts of a network — are looking for by generating examples.



**Attribution** [1] studies what part of an example is responsible for the network activating a particular way.

Source: Olah et al. (2017)

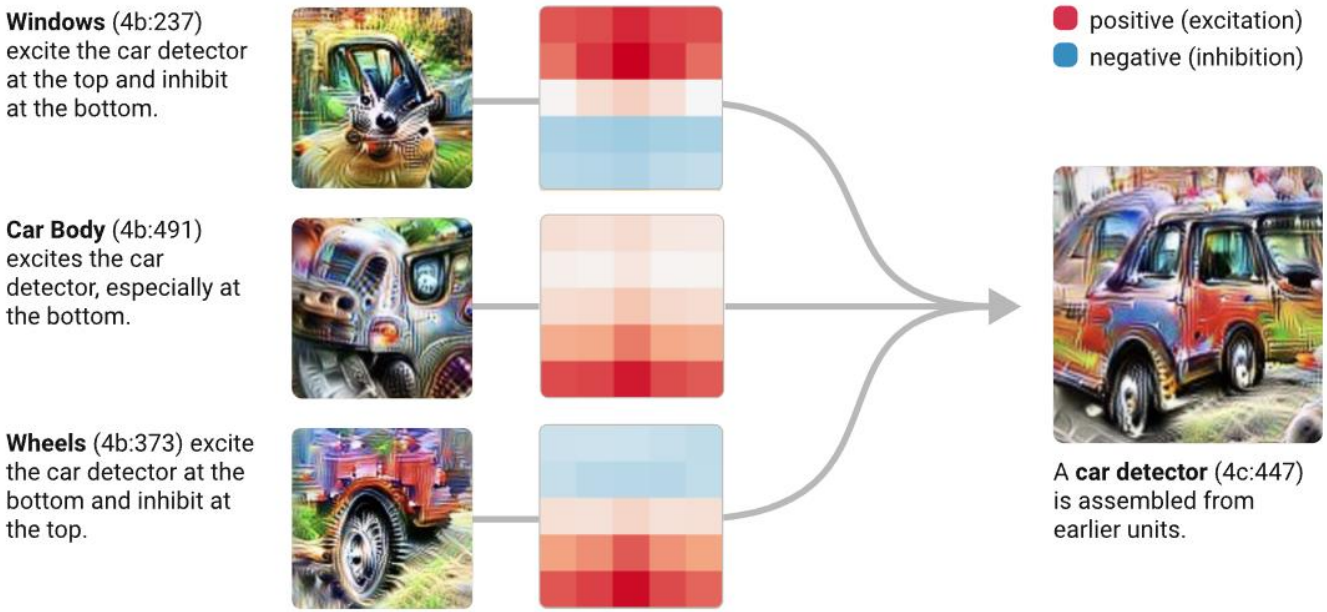
# Feature Relevance Explanation: Mechanistic Interpretability




Source: Olah et al. (2017)

# Feature Relevance Explanation: Mechanistic Interpretability

## Circuits




Source: Olah et al. (2020)

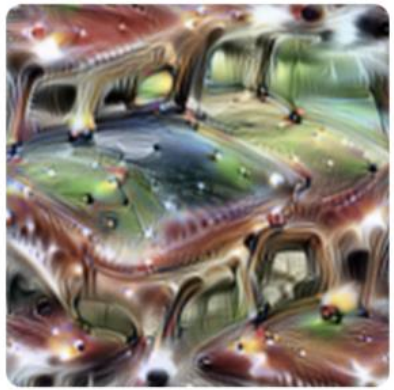
# Feature Relevance Explanation: Mechanistic Interpretability

Polysemanticity: even if we have an idea of what a neuron is doing…

…. we don‘t necessarily know **what else it might be doing**

For instance, one neuron activated by…

… cat faces…



… car hoods…

… and cat legs

Source: Olah et al. (2020)

# Feature Relevance Explanation: Mechanistic Interpretability

- Approach to attempt interpretation of a network's individual neurons:
  Compare them against dataset examples and generate inputs that activate them

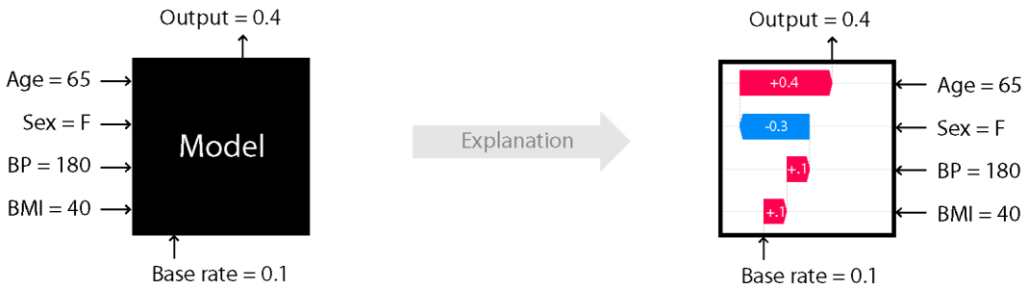
- Neurons form together into circuits which explain more complex behavior

- Do not forget polysemanticity: sometimes a neuron tracks multiple distinct features

- Similar research on large language models, e.g., Templeton, A. (2024)

# Interpretable / Explainable Machine Learning
# SHAP (SHapley Additive exPlanations)

SHAP



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# Audience Q&A

# Hands-on Exercise with SHAP

# Causal Inference

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# What is the fundamental problem of causal inference?

# Causal Inference and Potential Outcomes

- Correlation does not imply causation
- Understanding **cause and effect** is highly relevant for **decision-making**
  - Medical treatments
  - Impact of marketing campaigns
  - Impact of policy changes (e.g., new law, financial aid to firms and individuals, …)
- The **fundamental problem** of causal inference makes **direct observation impossible**
  - Additional assumptions required for estimation of causal effects

# Potential Outcomes

Causal effect: the magnitude by which an **outcome** is changed by an **interventional change**, between the real world and the counterfactual world

$$E\left[Y_{real,A=1}\right] - E\left[Y_{counterfactual,A=0}\right]$$

Real World: **do(T=1)**

Counterfactual World: **do(T=0)**

# Rubin Causal Model

1. Conditional independence ( or conditional ignorability/exogeneity or conditional unconfoundedness):

$$(Y_i^1, Y_i^0) \perp\!\!\!\perp D_i | X_i.$$

2. Stable Unit Treatment Value Assumption (SUTVA) (or counterfactual consistency):

$$Y_i = Y_i^0 + D_i(Y_i^1 - Y_i^0).$$

3. Overlap Assumption (or common support or positivity):

$$\forall x \in supp(X_i), \quad 0 < P(D_i = 1 | X_i = x) < 1,$$
$$P(D_i = 1 | X_i = x) \stackrel{\text{def}}{=} e(x). \tag{1}$$

4. Exogeneity of covariates:

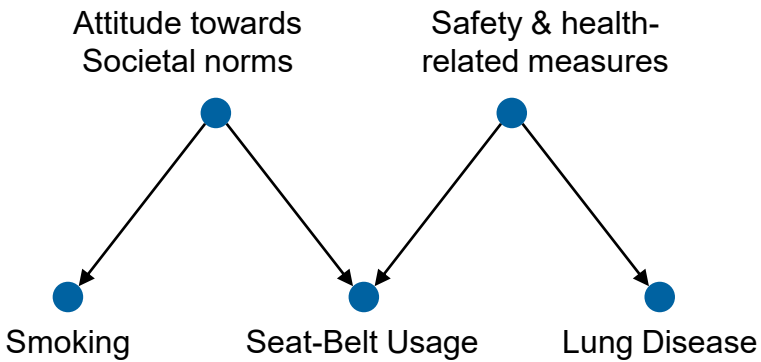
$$X_i^1 = X_i^0.$$

- Treatment **assignment** is **independent** of the **potential outcomes** (no “perfect doctor”)
- **No hidden variation** between **treated and control** groups
- No **subgroup** is **located entirely** within either the treatment or control group
- The **treatment** has **no effect** on the **other variables**

# Graphical Causal Models

# Graphical Causal Models

- **Directed Acyclic Graph** (DAG): help us to describe all relevant relationships




- "**Back-door criterion**" to determine which variables need to be included as confounders and which variables must be omitted
- **Causal Discovery**: attempt to automatically generate a DAG from the data
- **Causal Validation**: test a proposed DAG using (empirical) data

# Graphical Causal Models: Important Concepts

Chain

$$X \longrightarrow Z \longrightarrow Y$$

Fork

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ X & & Y \end{tikzcd}$$

Collider

$$\begin{tikzcd} Z \arrow[dr] & & X \arrow[dl] \\ & Y & \end{tikzcd}$$

When should you control for Z?

slido

# When should you control for Z?

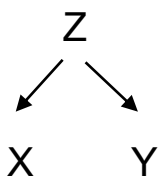
# Graphical Causal Models: Important Concepts

Chain

Exogenous variables: X
Endogenous variables: Z and Y
Z is a mediator

Do not control for Z

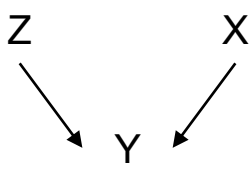
Fork




Exogenous variables: Z
Endogenous variables: X and Y

Control for Z (confounder)

Collider

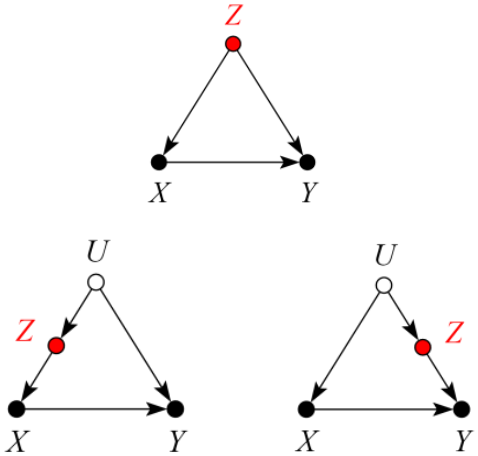



Exogenous variables: Z and X
Endogenous variables: Y

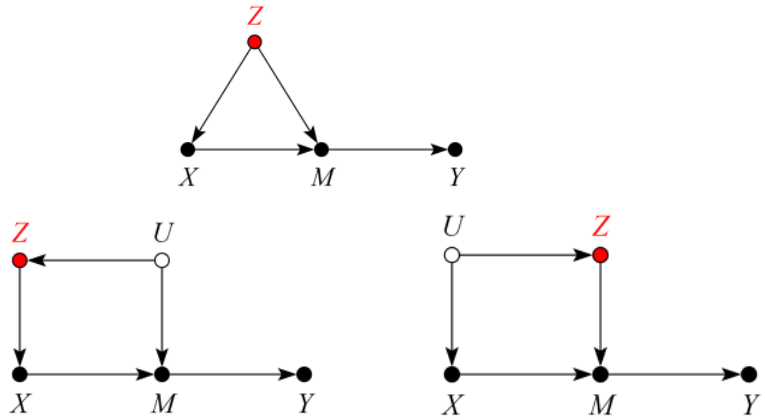
Do not control for Z

Source: Pearl, J., & Mackenzie, D. (2018).

# Graphical Causal Models: Good Controls



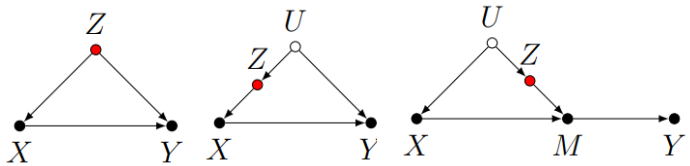

Control for Z
(block the **back-door** path from X to Y)




Control for Z
(block the **back-door** path from X to Y **through the mediator M**)

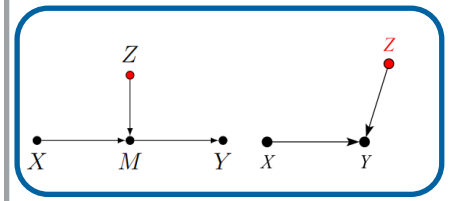
# Graphical Causal Models – Good, Neutral, and Bad Controls

## Good Controls

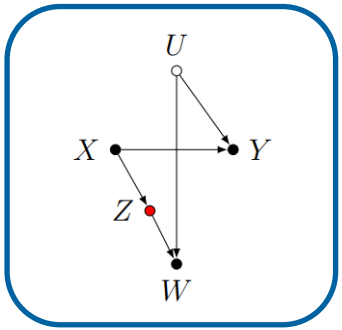


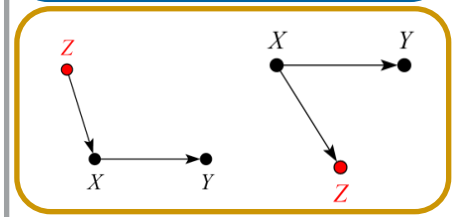

## Neutral Controls

Good for precision!



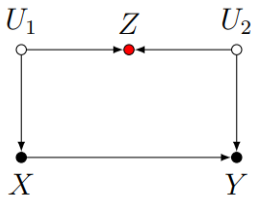

Possibly helpful in case of selection bias



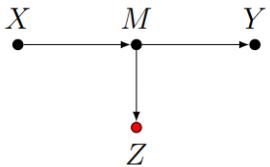




Possibly bad for precision

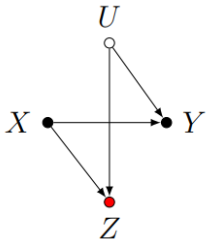
## Bad Controls



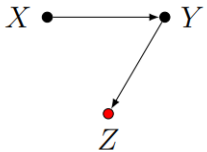

**M-Bias**
Z is a pre-treatment variable



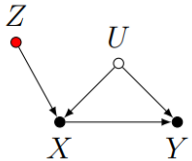

**Overcontrol bias**
~ partially controlling for M




**Selection bias**
opens the backdoor from U




**Case-control bias**
Z is effect of the outcome




**Bias amplification**

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# Audience Q&A

# Causal Trees and Causal Forests

# Causal Trees - Build Procedure: Double-Sample Trees

Input: $n$ training samples with features $X_i$, response $Y_i$ and treatment assignment $W_i$

Step 1: Divide the samples into a **splitting subsample** and an **estimation subsample**

Step 2: **Grow a tree** via recursive partitioning (with the **splitting subsample**)

–For each possible split: **estimate treatment effect** by comparing samples with and without treatment

–Use the **split that best optimizes the treatment effect** in the two new nodes

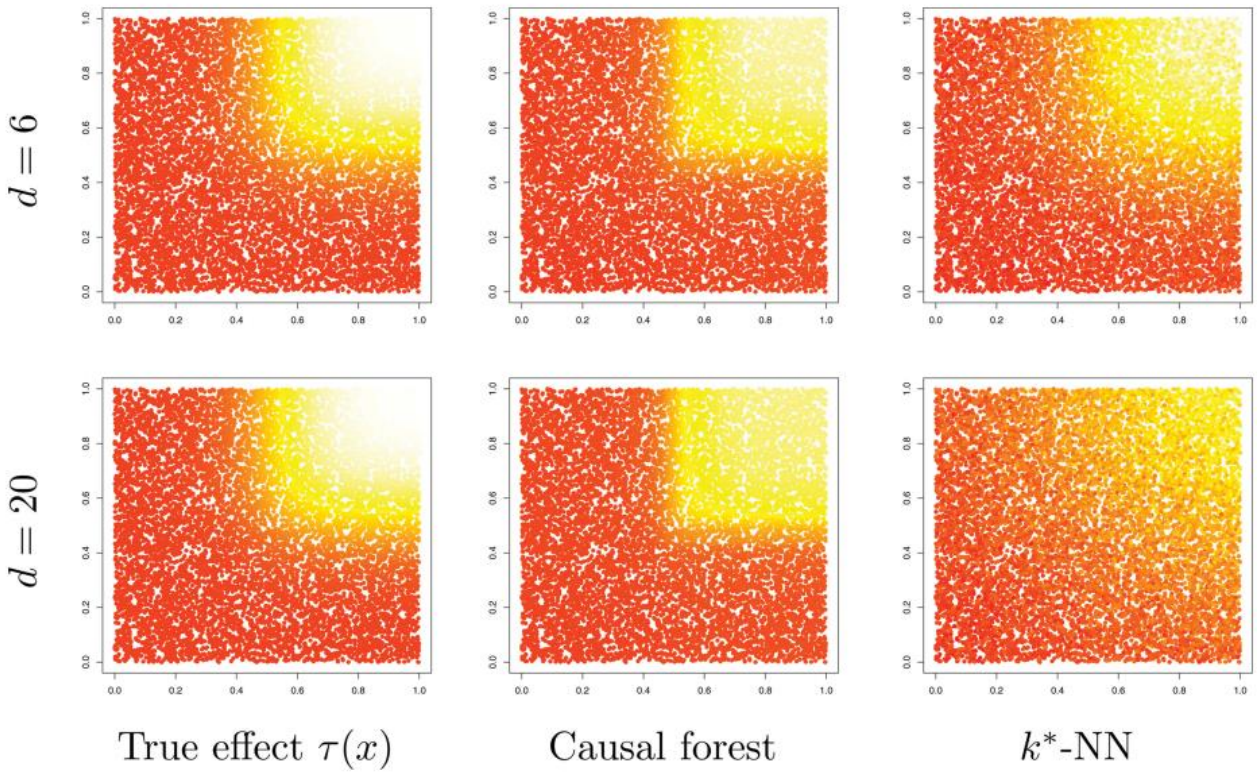
Step 3: Estimate **conditional treatment effect** on samples within leaf nodes (with the **estimation subsample**)

# Causal Forests

- Causal forests: random forests with causal trees (Athey and Wagner, 2019)
- Similarly: **splitting subsample** and **estimation subsample**

- Splitting criterion **optimizes for splits** associated with **treatment effect heterogeneity**
  - Find leaves where the **treatment effect is constant**, but **different from other leaves**
  - **Only split** when the nodes will still contain **enough samples for estimation**
- Estimate the conditional average treatment effect (CATE) using the leaves

# Causal Forests




Source: Wagner & Athey (2017)

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# Audience Q&A

# Hands-on Exercise with EconML and PyWhy

slido

# Which of the following are examples of regularization techniques? (Select all that apply)

slido

# What is transfer learning most useful for?

slido

# What does the bias-variance trade-off refer to?

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# Anomaly Detection

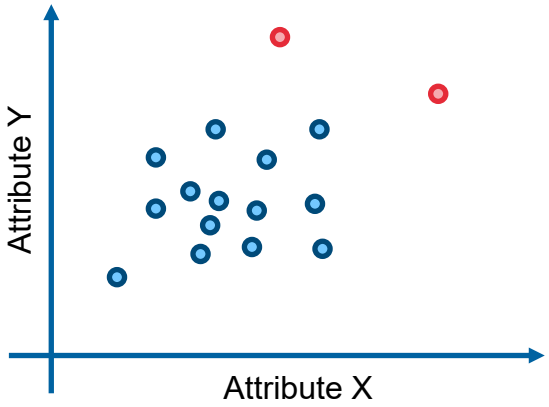
# Anomaly Detection: Agenda

- The foundations
  - What is anomaly detection?
  - Anomalies in various domains.
  - Three main problems it solves.
  - Anomaly types.
  - Challenges in Anomaly Detection.
  - Existing Models
    - Statistical models
    - Classical models
    - Deep Learning models
    - Foundation models
- Anomaly Detection in Finance
  - Use cases
- Best practices
  - What you should (not) do in anomaly detection.

# Anomaly Detection: Foundations

## What is an anomaly?

*„An observation which deviates significantly from the majority of the data or its distribution“*




Distance-based (math)

Feature-based (context)

# Anomaly Detection: Foundations

What is an anomaly?

*„An observation which deviates significantly from the majority of the data or its distribution“*

**anomaly ≠ crime**

It‘s a signal for further investigation

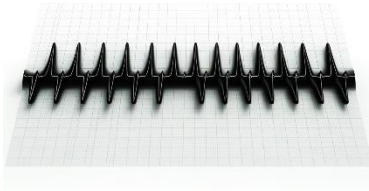
Distance-based (math)

Feature-based (context)

# Anomaly Detection: Foundations

## Anomalies in various domains

**Healthcare:** an ECG (Electrocardiogram) trace where every individual beats looks normal, but the **rhythm (sequence)** is irregular and life-threatening.



**Manufacturing:** a turbine that normally vibrates at frequency of 55Hz suddenly decreases to 50Hz every ten minutes. Such repeated „drift“ in vibration data indicates a turbine slowly disintegrating.

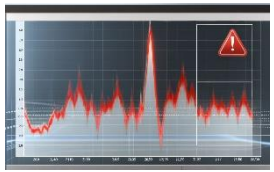
**Cybersecurity**: a user account typically uploads 10Mb suddenly uploads 100Gb to external IP at 2am. Login credentials are correct, but the time and location are highly suspicious.

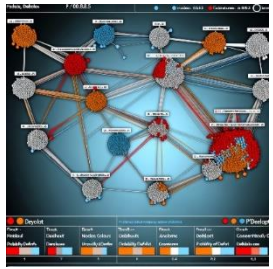
# Anomaly Detection: Foundations

## Anomalies in various domains

**Finance**

- **Tabular data**: a series of 50 transactions, each exactly €9.990, sent within 10 minutes. Could be designed to stay just below the €10.000 mandatory reporting threshold.
- **Time Series data**: a sudden massive drop in a single quarter of GDP. This is a **Global Shock** anomaly in Q2 2020 during COVID-19.
- **Graph data**: Bank A lends to Bank B, which lends to Company C, which uses that money to buy „toxic assets“ from Bank A. Such **Curcular Flow** is used to hide high probability of default loan and „augment“ the balance sheet.





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# Why anomaly detection is relevant?

# Anomaly Detection: Foundations

## The problem of large data volume

- Global transaction volume surpasses **billions a day**.
- Humans cannot audit the system. AD filters 99,9% of normal data so that human experts **focus of 0.1% that actually matter**.
- Advanced and optimized algorithms are necessary to find a **niddle in a haystack**.

# Anomaly Detection: Foundations

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## The problem of „(Un)Known Unknowns“

- Rule based systems are **bypassed by sophisticated criminals** (using AI).
- You know **something “fishy” is happening** (“Known”), but do not know how it looks like (“Unknown).
- AD looks for **deviations from normal behaviour** to catch new frauds topologies („unknown unknowns“)

# Anomaly Detection: Foundations

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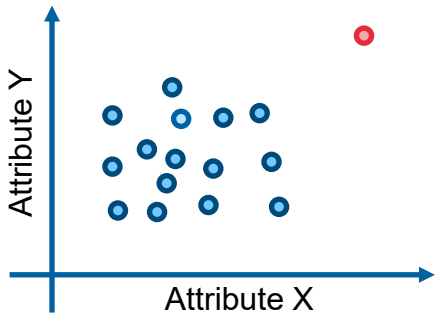
## The problem of hidden systemic risk

- Financial crises often starts as a **small „cracks“ in the data**; e.g. a shift on bank liquidity patterns or a „drift“ in bond market.
- AD is a **SupTech (supervisory tech) tool** for central bank.
- It monitors and detects the **first signs of a „black swan**“ event.

# Anomaly Detection: Foundations

## Point anomalies

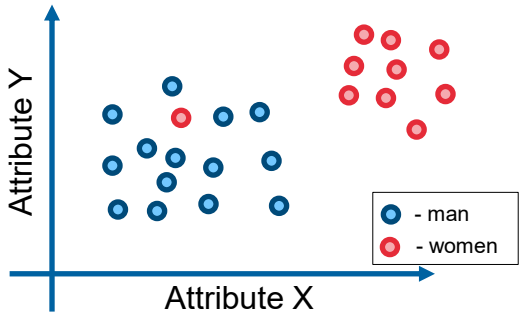
- **Definition**: A single data point that stands out as an extreme outlier from the entire distribution.
- **Example**: A $500M transaction from a retail account with a $5k average balance.
- **Note**: usually the "low-hanging fruit" for automated flagging.



# Anomaly Detection: Foundations

## Contextual anomalies

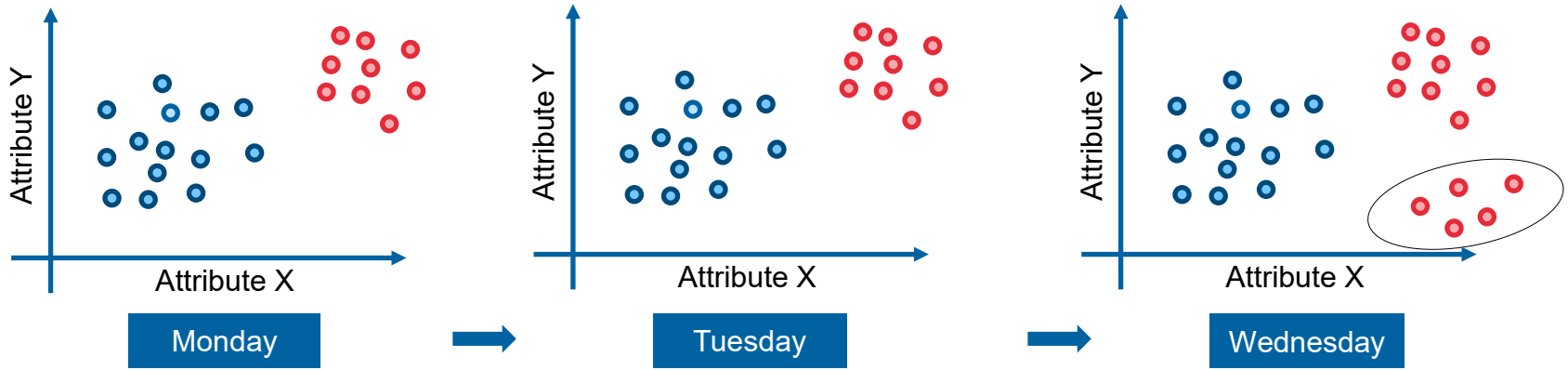
- **Definition**: Data that is "normal" in general, but anomalous given a specific context (time, season, or segment).
- **Example**: A surge in ATM withdrawals that is normal on payday, but anomalous on a random Tuesday.
- **Note**: require domain knowledge (e.g., knowing the local holiday calendar or market hours).



# Anomaly Detection: Foundations

## Collective anomalies

- **Definition**: A sequence or group of events that are suspicious only when viewed together.
- **Example**: Structuring (Smurfing): 100 separate $9k transfers to avoid the $10k reporting threshold.
- **Note**: the most difficult to detect and usually require advanced Graph Analysis or Sequence Modeling.



# Anomaly Detection: Foundations

## Challenge I – The Labeling Gap

- **Problem**: Absence of "Ground Truth" Data
- In financial domain, verified examples of "anomalous" cases are rare, confidential, or only identified months after the event.
- **Constraint**: Traditional Supervised Learning (which requires labeled "Good" vs. "Bad" examples) is often impossible to implement.
- **Solution**:
  - Reliance on Unsupervised Learning (detecting deviations from the norm).
  - Exploring Semi-Supervised models: Training the system strictly on "normal" historical data so it recognizes anything else as an outlier.

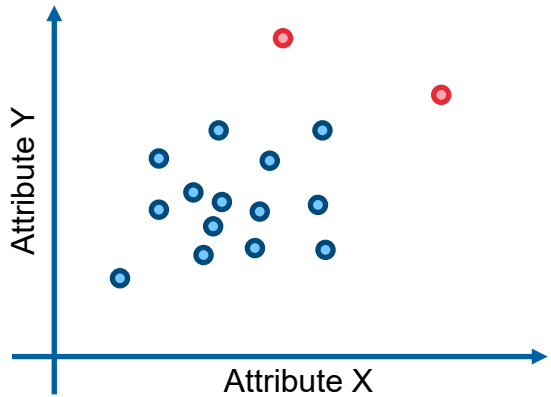
| | X1 | X2 | X3 | X4 | X5 | X6 | outlier |
|---|---|---|---|---|---|---|---|
| 1 | 3.99 | 1.21 | 2.07 | B | IT | M_42 | yes |
| 2 | 2.99 | 10.01 | 6.17 | S | GB | K_23 | yes |
| 3 | 0.0 | 2.09 | 3.43 | B | DE | K_32 | no |
| 4 | 11.09 | 0.08 | 9.41 | S | FR | M_52 | yes |
| 5 | 1.4 | 0.11 | 7.21 | S | IT | M_42 | no |

✓ ✗

# Anomaly Detection: Foundations

## Challenge II – The Imbalance Paradox

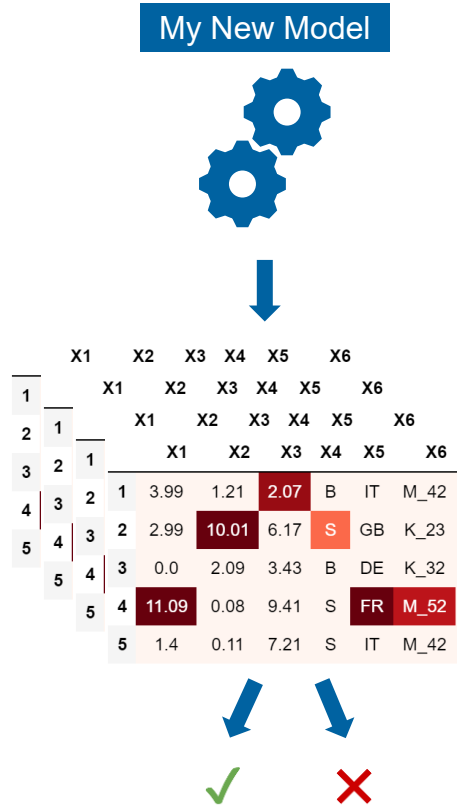
- **Problem**: Anomalies are the "0.1%"
  - Financial datasets are overwhelmingly "clean."
  - Anomalies are the needle in the haystack.
- **Trap**:
  - If 99.9% of transactions are legitimate, a broken model that flags everything as "normal" is still 99.9% accurate.
  - Accuracy is a misleading metric for financial stability and oversight.
- **Solution**:
  - Abandon "Accuracy" as a KPI.
  - Focus on Precision (avoiding "False Alarm") and Recall (the cost of missing a critical event).



# Anomaly Detection: Foundations

## Challenge III – The Benchmarking Vacuum

- **Problem**: No Universal "Gold Standard"
  - Unlike vision or NLP domains, there are limited, industry-wide labeled datasets for financial anomaly detection.
- **The Algorithm Dilemma**:
  - An algorithm that works for AML (Anti-Money Laundering) may fail for Liquidity Stress detection.
  - Success is highly context-dependent; "Plug-and-Play" solutions do not exist.
- **Solution**:
  - Develop Institutional Benchmarks: Defining success based on specific central bank mandates.
  - Focus on Custom Validation Frameworks rather than off-the-shelf software claims.



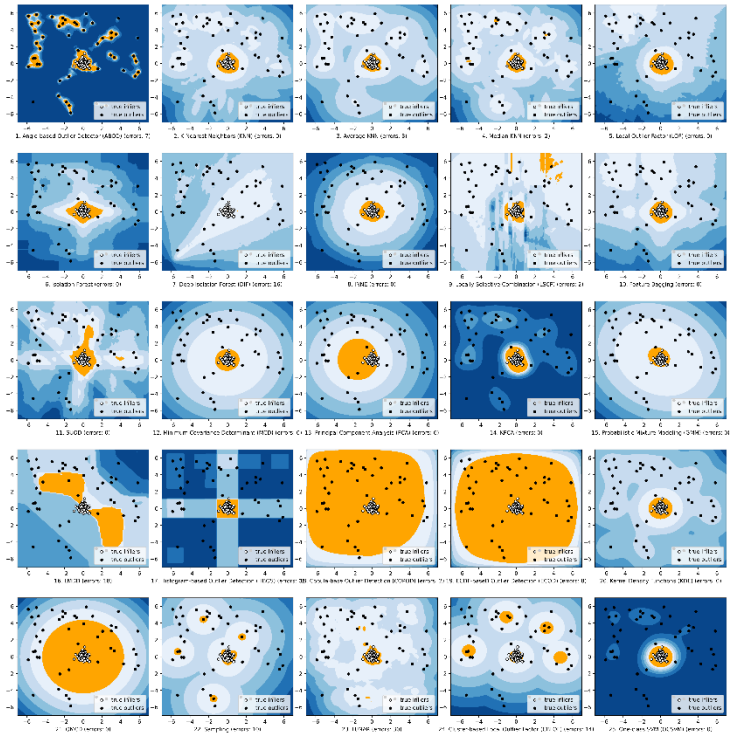
# Anomaly Detection Models

# Anomaly Detection: Models

## Anomaly Detection Family

Deep SVDD
DBSCAN
XGBOD
PCA
VAE
ABOD
SOS
IForest
Autoencoder
COF
ECOD
CBLOF
KNN
HBOS
KPCA
LOF
COPOD
OC-SVM
AnoGAN
LMDD
MCD
MO-GAAL
MAD
LOCI
SOD
GMM

25 outlier detection algorithms on synthetic data



PyOD library https://pyod.readthedocs.io/en/latest/index.html

# Anomaly Detection: Models

Evolution of Anomaly Detection field:

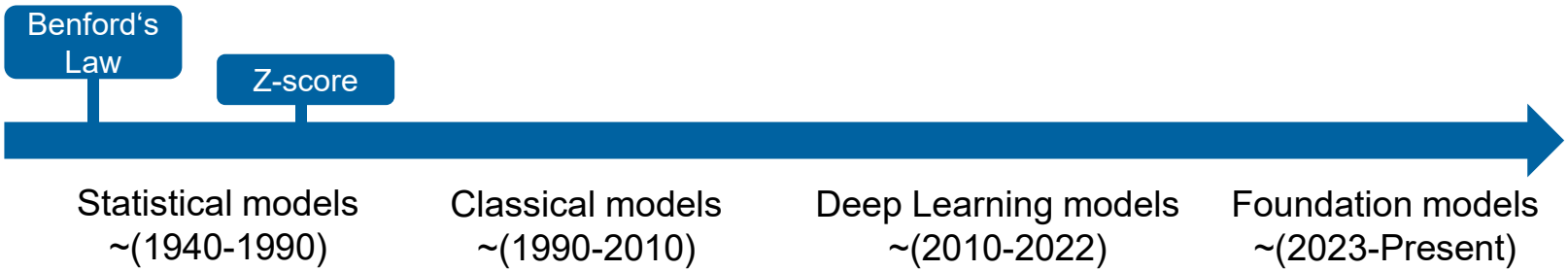
1. **Statistical models**: rely on predefined thresholds and probability distributions, static rules.
2. **Classical (ML) models**: classical ML techniqies, shift to adaptive algorithms.
3. **Deep Learning models**: neural networks, designed to capture non-linear patterns in large datasets.
4. **Foundation models**: large-scale pre-trained models, can be adapted for diverse downstream tasks.

| Statistical models ~(1940-1990) | Classical models ~(1990-2010) | Deep Learning models ~(2010-2022) | Foundation models ~(2023-Present) |
|---|---|---|---|

# Anomaly Detection: Models

**Statistical models**

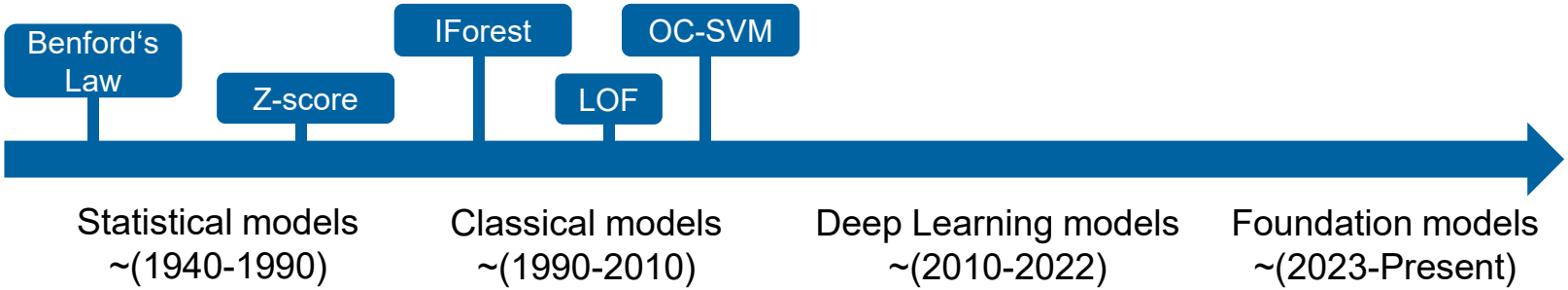
- **Benford‘s Law**
  - Predicts the frequency of leading digits in natural datasets (like ledger entries)
  - Frequenctly used by auditors to detect financial frauds.
- **Z-score /IQR**
  - Measures how many standard deviations a data point is from the mean.
  - The 3-sigma rule usually is used to detect deviations from historical data.



# Anomaly Detection: Models

**Classical Models**

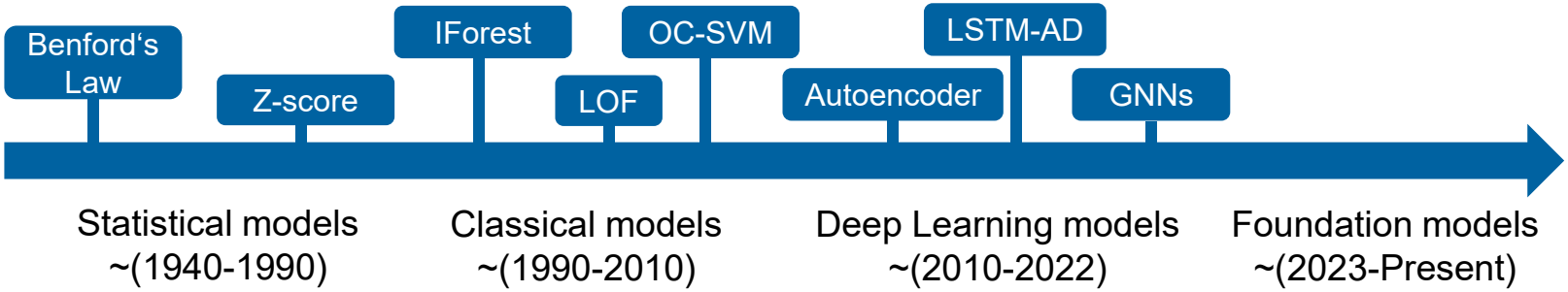
- **Isolation Forest**
  - Uses random trees to isolate anomalies.
  - Remains industry standard, very powerful and efficient on high-dimensional tabular data.
- **LOF**
  - Measures the local density deviation of a point compared to ist neighbors.
  - Great for finding „clusters“ of fraud. Could be computationally expensive.
- **OC-SVM**
  - Learns a „boundary“ around normal data, anything outside is an anomaly.
  - Can become computationally expensive on large datasets.



# Anomaly Detection: Models

**Deep Learning models**

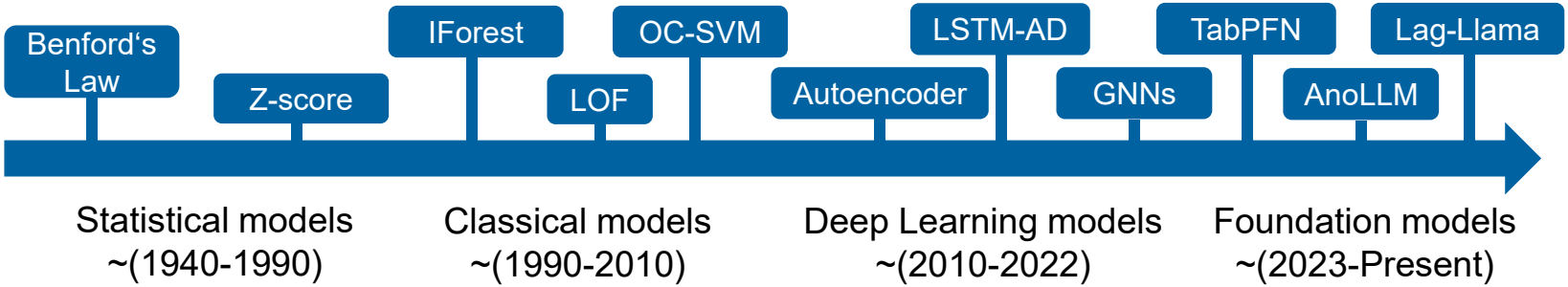
- **(variational/denoising) Autoencoder**
  - A neural network, that learns to compress and reconstruct data.
  - If the reconstruction of a sample fails it is likely an anomaly.
- **LSTM-AD (Long Short-Term Memory)**
  - Designed for time-series. The model learns a „normal“ sequences and flags deviations in the pattern.
- **Graph Neural Networks (GNNs)**
  - Finding the anomalies in the relationships between entities. Essential in Anti-Money Laundering (AML)



# Anomaly Detection: Models

**Foundation models**

- **TabPFN**
  - A Transformer pre-trained on millions of synthetic tabular datasets.
  - Uses In-Context Learning (ICL), no training is required.
- **AnoLLM**
  - Treats tabular data as language serializing rows of data into text of strings.
- **Lag-Llama**
  - Foundation model for probabilistic time-series.



# Anomaly Detection in Finance

# Anomaly Detection: Use Cases

**Financial Stability & Systemic Risk Monitoring**

1. **Liquidity Stress Detection:** Identifying unusual patterns in systemic liquidity, such as a sudden drop in interbank lending that could signal an upcoming market shock.
   Example: "*Predicting bank distress in Europe*", Despo Malikkidou and Wolfgang Strohbach, 2025
   Staff Papers - Predicting bank distress in Europe.pdf
2. **Early Warning Systems (EWS):** analyze bank balance sheets and sentiment in corporate text (via NLP) to identify idiosyncratic risks that could become systemic.
   Example: "*SAFE: An early warning system for systemic banking risk*", Mikhail V. Oet et.al 2013
   SAFE: An early warning system for systemic banking risk
3. **Interconnectedness Analysis (Graph):** Detecting anomalies in the network of exposures between banks and non-bank financial intermediaries to limit the propagation of shocks.
   Example: "*The network topology of CHAPS Sterling*", Becher et.al , 2008
   The network topology of CHAPS Sterling

# Anomaly Detection: Use Cases

**Anti-Money Laundering (AML)**

1. **Real-Time Payment Monitoring:** Detecting unauthorized activity or fraudulent deposits in real-time rails before funds are released.
   Example: "*Project Aurora: The power of data, technology and collaboration to combat money laundering*", BIS Innovation Hub, 2023 Project Aurora: the power of data, technology and collaboration to combat money laundering across institutions and borders
2. **Cross-Border Illicit Flow Detection:** Identifying suspicious transaction patterns across international corridors, particularly those involving digital currencies.
   Example: "*Illicit Financial Flows: Concepts, Measurement, and Evidence*", Matthew Collin, 2019 Illicit Financial Flows: Concepts, Measurement, and Evidence | The World Bank Research Observer | Oxford Academic

# Anomaly Detection: Use Cases

**Monetary Policy & Data Quality Assurance**

1. **Statistical Data Validation:** Automatically identifying reporting errors, outliers, or missing entries in the massive volumes of financial microdata collected from credit institutions.
   Example: "*Machine learning-based approaches for automatic data validation and outlier control of loan microdata in the Bank of Russia*", Dmitrii Diachkov, 2021 Machine learning-based approaches for automatic data validation and outlier control of loan microdata in the Bank of Russia
2. **Economic Indicator Forecasting:** Spotting deviations in traditional macro-financial conditions (e.g., credit growth, asset prices) that fall outside expected thresholds, which may influence interest rate decisions.
   Example: "*Unsupervised Learning for Anomaly Detection in Financial Markets and Crisis Prediction*", Akanbi Caleb, 2022 (PDF) Unsupervised Learning for Anomaly Detection in Financial Markets and Crisis Prediction

# Anomaly Detection: Base Practice

What you should and should not do.

1. **Baseline**: start with simple model (this is your baseline), then improve it with model complex.
2. **Human-in-the-loop**: usually you do not have labels, anomalies are rare-> domain expertise is essential (economists/auditors) to validate „hits“.
3. **Evaluation metrics**: no accuracy, but precision, recall, f1, PR-AUC.
4. **Iteration**: use the feddback from domain experts to „teach“ the model what a „FP“ looks like.
5. **Explanation**: XAI methods, dashboards, graphs are necessary tools to answer questions like „why this is an anomaly“.

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# Audience Q&A

# Hands-on Exercise

# References

Despo Malikkidou and Wolfgang Strohbach (2025), Predicting bank distress in Europe.

Mikhail V. Oet et.al (2013), SAFE: An early warning system for systemic banking risk.

Becher et.al (2008) The network topology of CHAPS Sterling.

BIS Innovation Hub (2023). Project Aurora: The power of data, technology and collaboration to combat money laundering.

Matthew Collin (2019) Illicit Financial Flows: Concepts, Measurement, and Evidence.

Dmitrii Diachkov, (2021) Machine learning-based approaches for automatic data validation and outlier control of loan microdata in the Bank of Russia.

Akanbi Caleb (2022), Unsupervised Learning for Anomaly Detection in Financial Markets and Crisis Prediction.

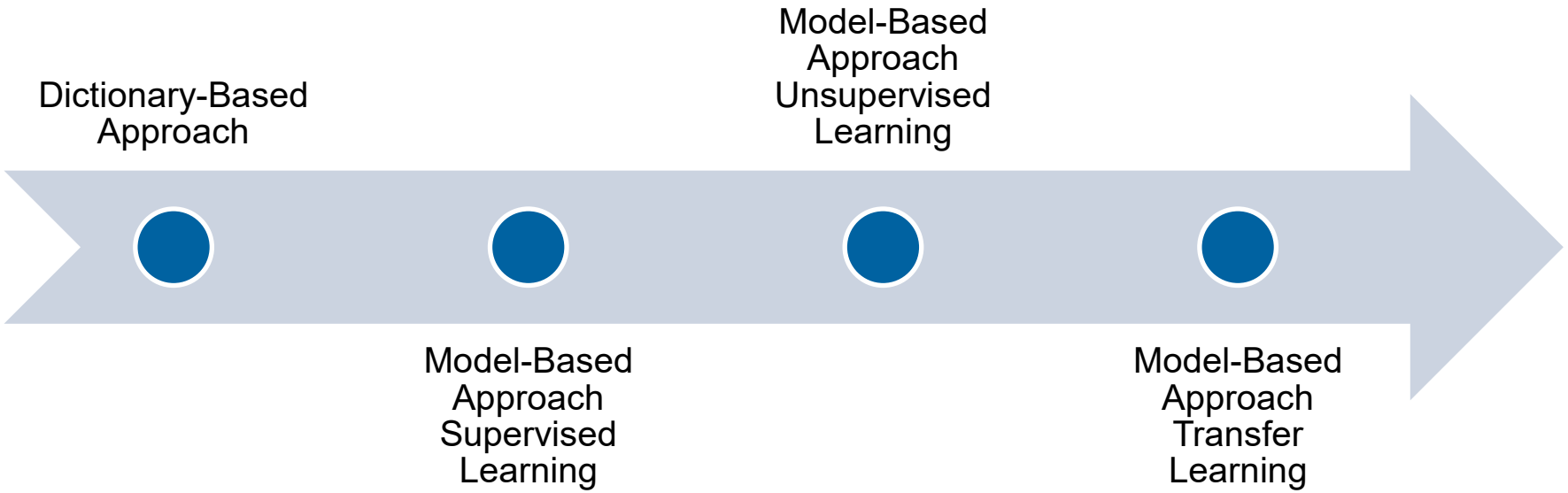
# Natural Language Processing

# Overview of NLP Tasks – Models Available on Hugging Face

| | | | | | |
|---|---|---|---|---|---|
| Text Generation<br>127,506 models | Text Classification<br>66,799 models | Token Classification<br>19,215 models | Fill-Mask<br>12,838 models | Question Answering<br>11,907 models | Feature Extraction<br>9,682 models |
| Sentence Similarity<br>5,806 models | Translation<br>4,525 models | Summarization<br>1,885 models | Zero-Shot Classification<br>291 models | Table Question Answering<br>103 models | |

Source: https://huggingface.co/

# Natural Language Processing Timeline



# Dictionary-Based Approach

- Bag-of-words model (unigrams)
- Focus on term frequencies (order does not matter)
- Pre-process text (remove stop words, remove unwanted characters, tokenize/lemmatize, …)

"My model is the best"

Document

Corpus

Document Term Matrix

| | bank | bad | fees | good | service |
|---|---|---|---|---|---|
| Document 1 | 1 | 0 | 1 | 0 | 0 |
| Document 2 | 3 | 0 | 2 | 1 | 0 |
| Document 3 | 0 | 0 | 0 | 1 | 1 |

- For sentiment analysis – need a dictionary of positive and negative words

**Tone =** **-**

# Dictionary-Based Approach

- Tf-idf: terms **weighted** by their **importance** in the document and in the corpus
  - Term frequency: $Tf_{w,D} = \frac{\text{Number of times } \textbf{word w} \text{ appears in } \textbf{document D}}{\text{Length of } \textbf{document D}}$
  - Inverse Document frequency: $Idf_{w,C} = log\left(\frac{\text{Number of documents in } \textbf{corpus C}}{\text{Number of documents containing } \textbf{word w}}\right)$
- Effective if opinions are **explicit and straightforward**
- Cannot handle implicit opinions, metaphors, sarcasm and more **complex language**: “Bought the couch a month ago and a valley appeared today”

# Model-Based Approach - Supervised Learning

- Feature extraction from text
- Corpus → Document-Term-Matrix
- **Independent variables**: word frequencies
- **Dependent variable**: Document sentiment or other **labels** you would like to predict
- Need (a lot of!) **labeled data for training**: documents for which you know the “correct answer”

| | bank | bad | fees | good | service |
|---|---|---|---|---|---|
| Document 1 | 1 | 0 | 1 | 0 | 0 |
| Document 2 | 3 | 0 | 2 | 1 | 0 |
| Document 3 | 0 | 0 | 0 | 1 | 1 |

# Model-Based Approach - Unsupervised Learning

Infer **underlying relationships** from the data

- **Words** that appear **together often** → Probably belong to the same topic
- **Documents** use **similar words** → Probably belong to the same topic

Probabilistic Latent Semantic Analysis (pLSA)
Hofmann (1999)

- **Topics** are latent variables from a multinomial distribution over observed **word** frequencies
- Restricted to **observed words** (“black swan” paradox)

Latent Dirichlet Allocation (LDA)
Blei et al. (2003)

- **Documents** are distributions over **topics**
- **Topics** are distributions over **words**
- Probability of **non-observed** word is small, but **not zero**

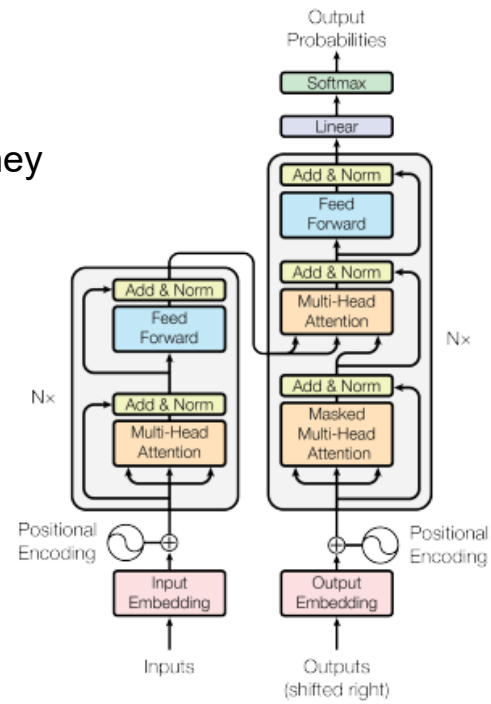
# Model-Based Approach - Transfer Learning with Transformers

**Transformers**

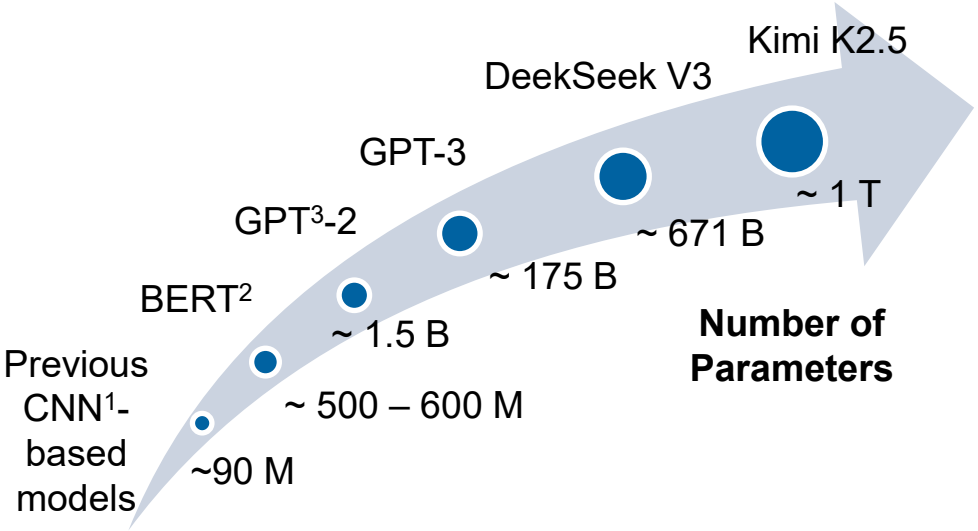
- Seminal paper “Attention is all you need” (Vaswani et al., 2017)
- Attention mechanism – learn context
- Track relationships in sequential data (e.g., words in a sentence), also if they are far apart (short and long-distance relationships)
- Find patterns – no need for labelled data

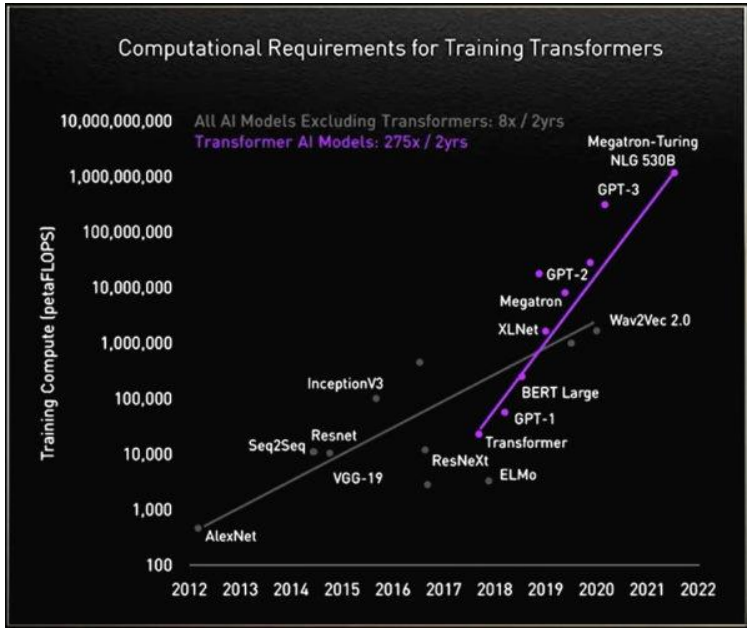
**Transformers for Text Mining**

- General-purpose unsupervised (language) models **pre-trained on large unlabeled corpora**
- **Richer and more accurate** representation of language **contextual relationships**
- **Fine-tune** with domain-specific data for several tasks (e.g., information retrieval, sentiment analysis, answering questions, …)



# Comparison of Transformer-Based Natural Language Processing Models



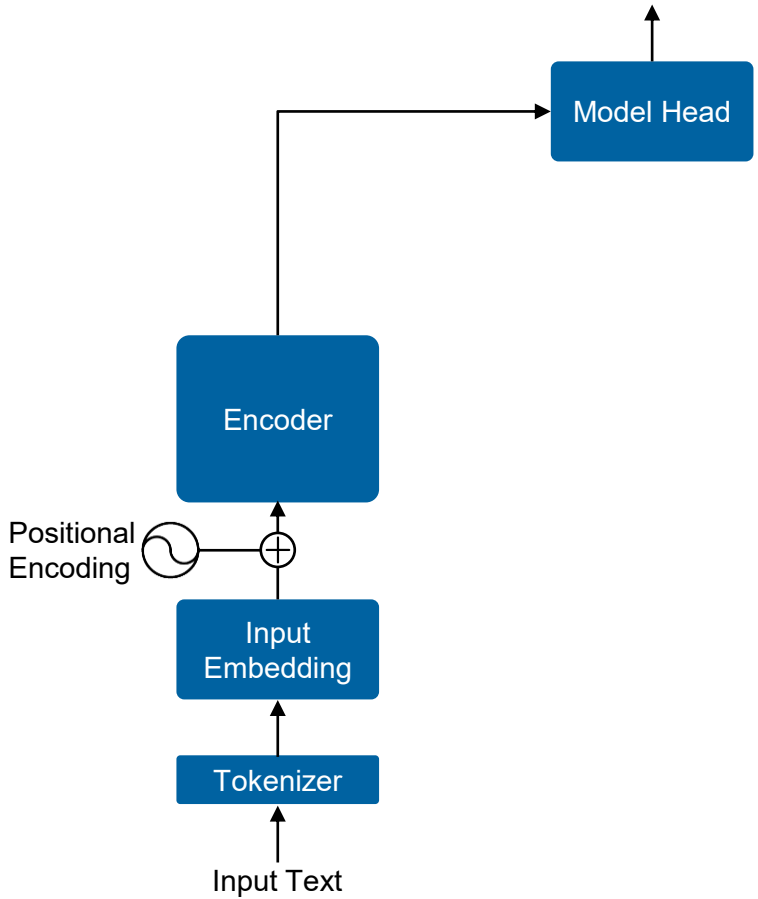




Source: NVIDIA (2022)

[1]CNN: Convolutional Neural Network (no transformer!)
[2]BERT: Bidirectional Encoder Representations from Transformers
[3]GPT: Generative Pre-Training

# Simplified Transformer Model Architecture: Encoder Only

- Great for transfer learning
- Examples:
  - BERT (Devlin et al., 2018)
  - RoBERTa (Liu et al., 2019)
  - DistilBERT (Sanh et al., 2019)



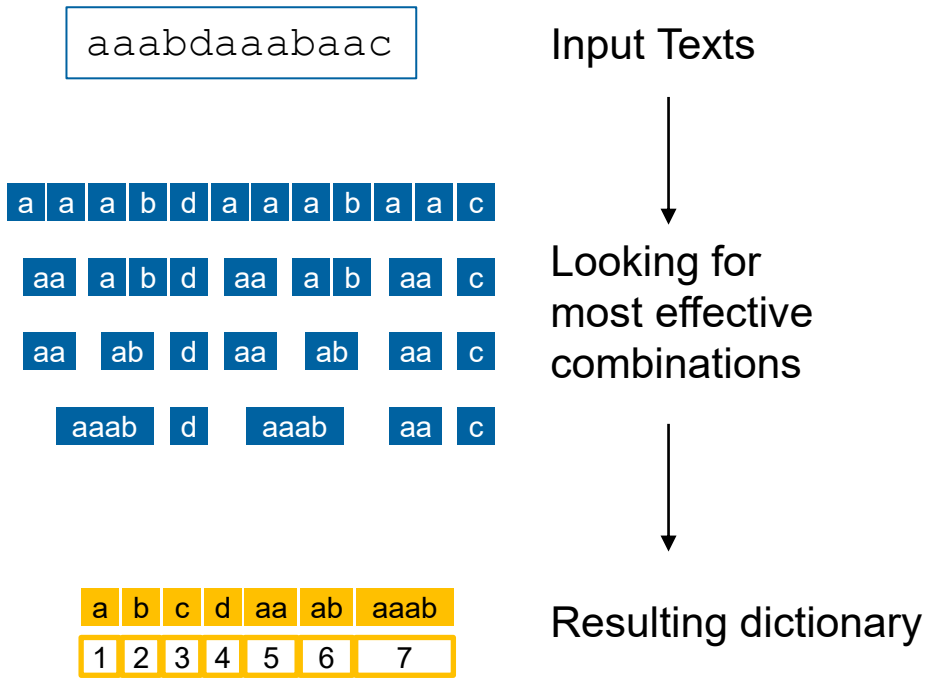
# Tokenization

## During Application

| Tokens | Characters |
|---|---|
| 162 | 993 |

To prevent jurisdictions from pursuing such policies at the expense of others, implementation monitoring is being intensified at the international level. By means of peer pressure and transparency, rigorous monitoring can contribute to ensuring a global level playing field. The results of numerous monitoring exercises are summarised in regular scoreboard and progress reports to the G20. In addition, international standard-setting bodies are stepping up their implementation monitoring instruments. To better coordinate the various mechanisms, the Financial Stability Board, or FSB for short, and the relevant standard-setting bodies have jointly developed a coordinating framework. Within this framework, six areas have been designated as deserving priority implementation and particularly intensive monitoring. I will now briefly address four of these priority areas - leaving out those dealing with SIFIs as another round table today is dedicated specifically to the regulation of SIFIs.

## During Training



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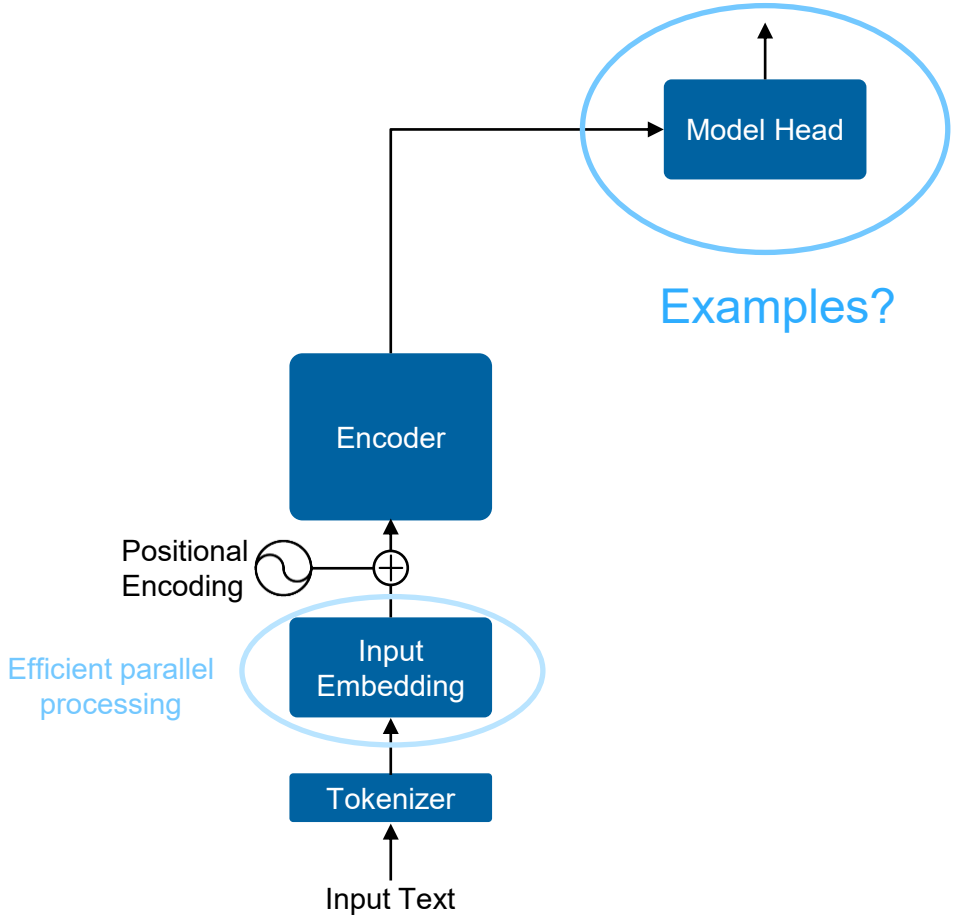
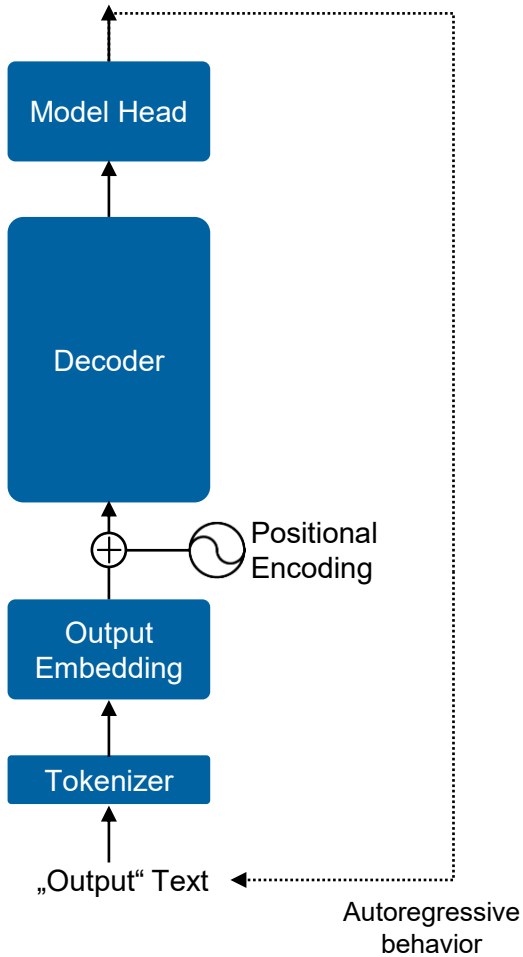


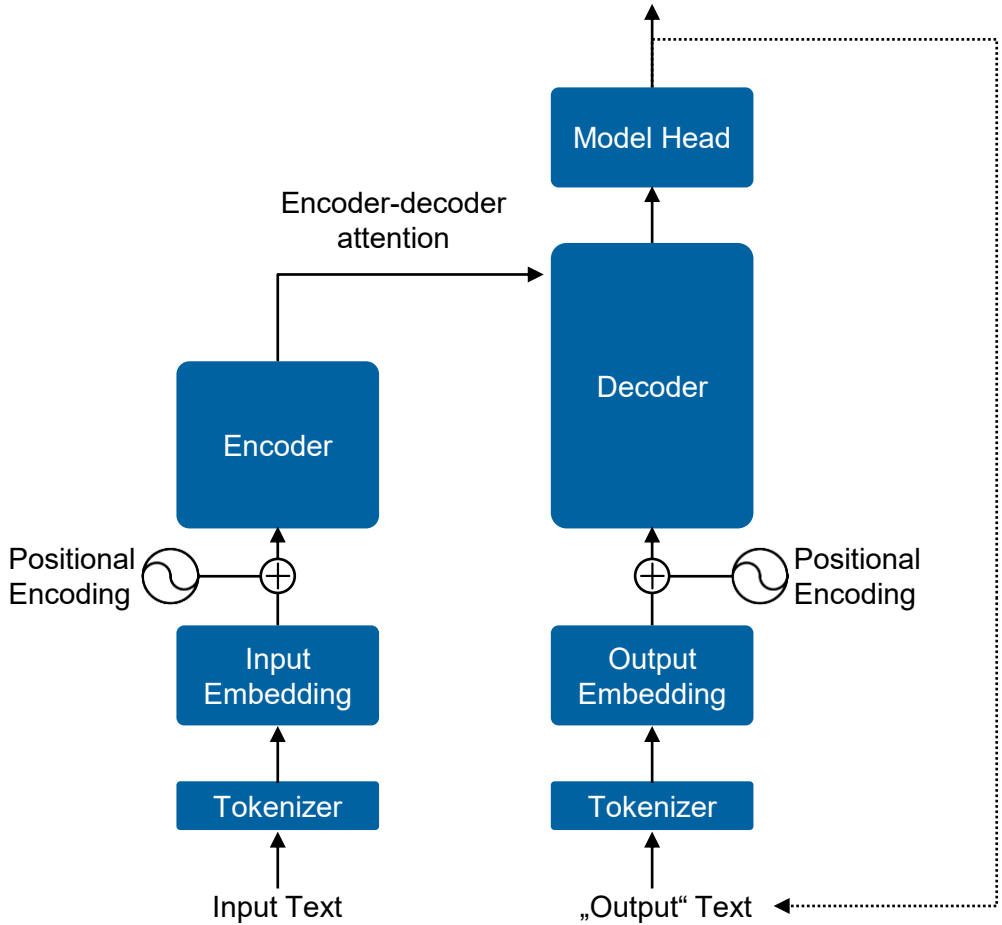
# Simplified Transformer Model Architecture: Decoder Only

- Focus on auto-regressive language modeling
- Predict next word in a sentence: good for text generation
- Examples:
  - GPT (Radford et al., 2018)
  - Gshard (Lepikhin et al., 2020)



# Simplified Transformer Model Architecture: Encoder and Decoder

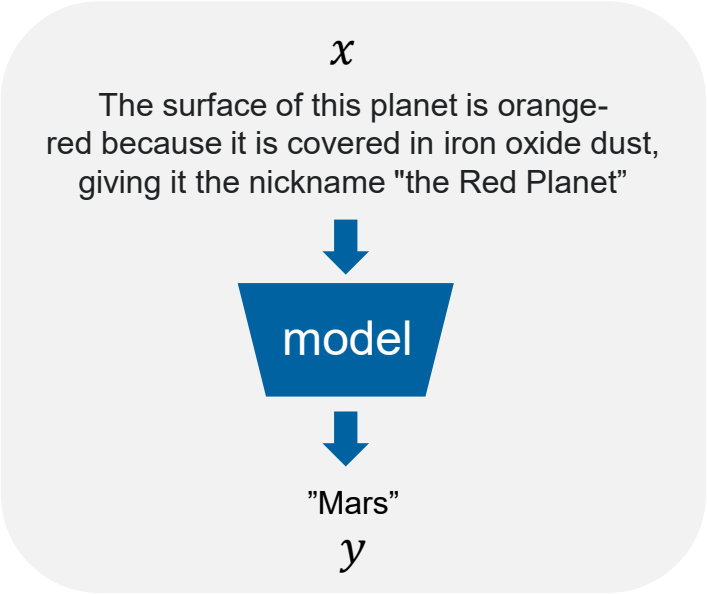
- Full encoder-decoder architecture
- Great for translation or summarization
- Examples:
  - Seminal paper (Vaswani et al., 2017)
  - T5 (Raffel et al., 2020)



# Model-Based Approach – Generative Modelling

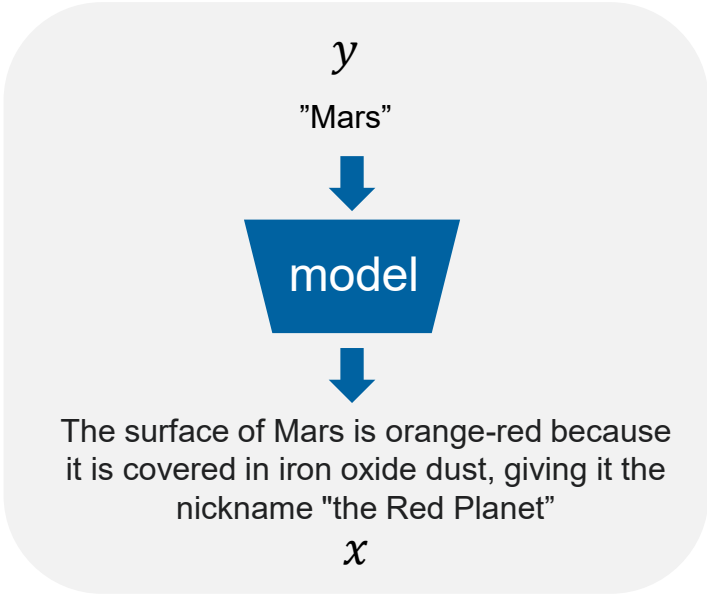
**Discriminative** Models

- Sample x → label y
- Single output



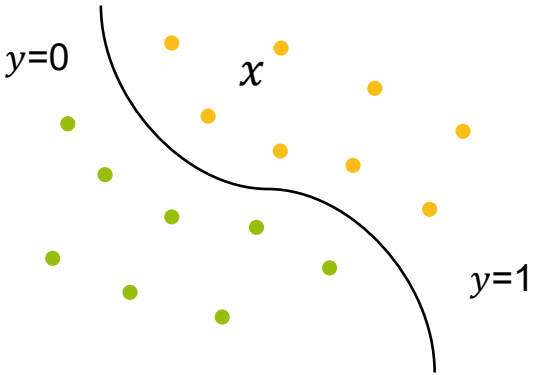

**Generative** Models

- Label y → sample x
- Many possible outputs

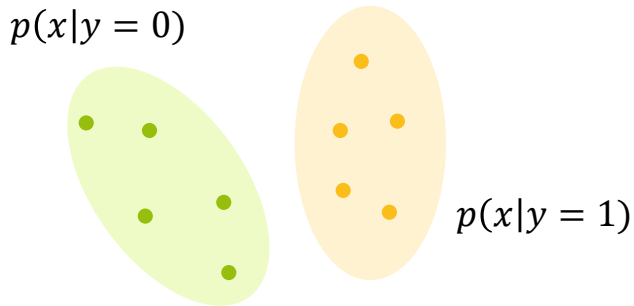


# Model-Based Approach – Generative Modelling

**Discriminative** Models $p(y|x)$

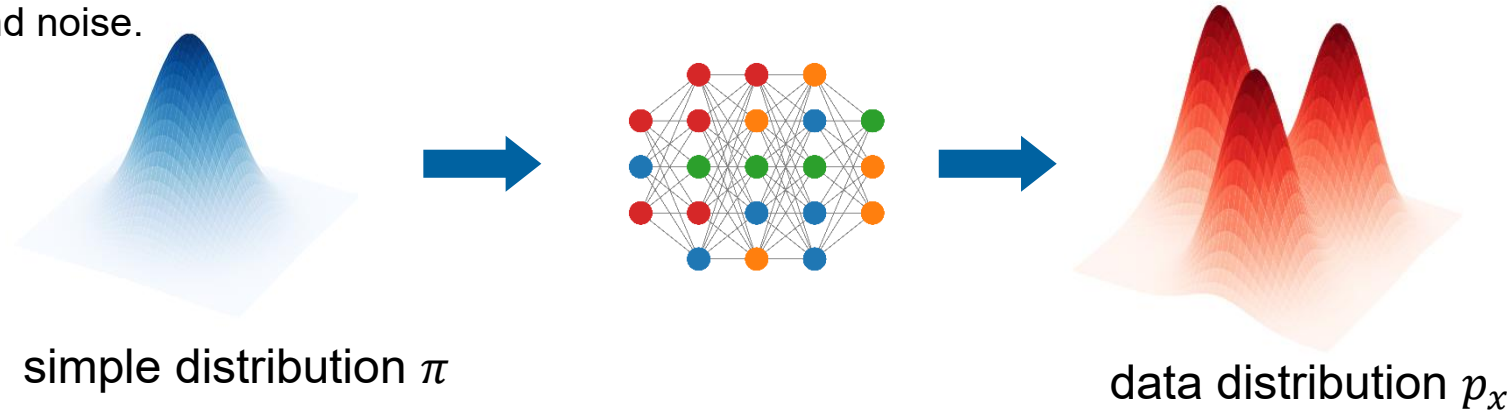



**Generative** Models $p(x|y)$



# Model-Based Approach – Generative Modelling

- **Variational Autoencoder** (Kingma et al. 2014): probabilistic latent space via encoder decoder.
- **Generative Adversarial Networks** (Goodfellow et al. 2014): generator-discriminator competition.
- **Autoregressive Models** (Van den Oord et al 2016): sequential modelling of conditional dependencies.
- **Diffusion Models** (Ho er al. 2020): learn to generate data by reversing gradual noise corruption.
- **Flow Matching** (Lipman et al. 2023): invertible data transformation for mapping between data and noise.



# Model-Based Approach – Generative Modelling

Natural Language Conversation

What are generative models?

Generative models are a class of machine learning models that learn to generate new data samples similar to those in a given dataset. Instead of just making predictions (as in discriminative models), they capture the underlying data distribution and can generate new, synthetic examples.

**Types of Generative Models**

1. **Probabilistic Models**
   - **Bayesian Networks**: Use probability distributions and conditional dependencies.
   - **Hidden Markov Models (HMMs)**: Used for sequential data modeling.
2. **Latent Variable Models**
   - **Variational Autoencoders (VAEs)**: Use neural networks to encode and decode data through a probabilistic latent space.
   - **Generative Adversarial Networks (GANs)**: Consist of a generator and a discriminator competing to create realistic data samples.
   - **Diffusion Models**: Gradually add noise to data and then learn to reverse the process to generate new samples.

$y$: prompt

$\mathrm{x}$: model output

# Model-Based Approach – Generative Modelling

Natural Language Conversation

What are generative models?

$y$: prompt

How can we learn such complex data modality?

- **Hidden Markov Models (HMMs)**: Used for sequential data modeling.

2. **Latent Variable Models**

- **Variational Autoencoders (VAEs)**: Use neural networks to encode and decode data through a probabilistic latent space.
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- **Diffusion Models**: Gradually add noise to data and then learn to reverse the process to generate new samples.

$x$: model output

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# Audience Q&A

# Hands-on Exercise with Transformers

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# Which of those are examples of suitable heads using textual data?

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# Which of those are examples of suitable heads, outside of NLP?

# Orientation for the Use Case Development

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# What is the main topic of the use-case you would like to work on during the on-site sessions?

# One Pager

- Contact Information
- Short Title
- Primary Application Area
- Objective
- Status Quo
- Data Description
- Stakeholders
- Proposed Methodology
- Potential Challenges and Assumptions

DEUTSCHE BUNDESBANK
EUROSYSTEM

**International Central Banking Courses**
**13560 - Introduction to Machine Learning**

**Contact Information**

**Name**
**Institution** **Email**

**Use Case Overview**

| | |
|---|---|
| **Short Title** | Please provide a short title for your use case that we can use to identify it. |
| **Primary Application Area** | e.g., Financial Stability, Banking Supervision, Payment Systems, Statistics, etc. |
| **Objective** | What is the main goal you want to achieve with this use case? (e.g., improve forecasting models, enhance regulatory compliance analysis, use unstructured data to improve existing analyses, etc.) |
| **Status Quo** | Please briefly describe the current state of the process or problem that you aim to address. Which methods do you currently use? What challenges do you face? |
| **Data Description** | What data are currently available? Include the type, size, and format of the data. Are there any challenges associated with the data (e.g., quality, availability, privacy concerns)? Do you need to acquire additional data (e.g., via web scraping)? |
| **Stakeholders** | Who are the key stakeholders involved? (e.g., departments, decision makers) |
| **Proposed Methodology** | Which methods from the course are you planning to apply? (e.g., Interpretable ML, Causal Inference, NLP) |
| **Potential Challenges and Assumptions** | What challenges do you foresee in implementing this use case? (e.g., data limitations, computational resources). What assumptions are you making about the data, methods, or outcomes? |

# Brief Overview of the Workshop Format in Frankfurt

### Pitch your idea

One-pager provides a rough structure for you to prepare your use case pitch for the first in-person day (~ **3 minutes** per person, following brief discussion)

### Active participation

We will alternate between **presenting** your ideas and early results, with break group **discussions** (promote synergies with similar use cases) and focused **coaching** sessions

### Required resources

If your use case relies on **internal data**, please make sure to have the data sets **accessible** and the analysis environment **installed** in **your own machine**, so that you can make the most of the coding sessions.

| IN-PERSON Session (Part 2) | | | |
|---|---|---|---|
| | **Tuesday**<br>**25 March 2025** | **Wednesday**<br>**26 March 2026** | **Thursday**<br>**27 March 2025** |
| 9:00 | **Welcome**<br>*Schwarz* | **Use-Case Sprint II**<br>Programming and Collaboration | **Open Coaching Session**<br>Individual and Group Coaching |
| 9.30 | **Introduction to the Workshop**<br>*Alves Werb, Seltmann, Bender DSZ* | *Course participants*<br>*Alves Werb, Seltmann DSZ* | *Course participants*<br>*Alves Werb, Seltmann DSZ* |
| 10.30 | *Break* | *Break* | *Break* |
| 10.45 | **Overview of Use-Cases**<br>Use-case Pitches by Participants<br>Guided Discussion in Breakout Groups<br>*Course participants*<br>*Alves Werb, Seltmann DSZ* | **Feedback Coffee**<br>Rotating Exchange and Feedback<br>*Course participants*<br>*Alves Werb, Seltmann DSZ* | **Short Presentations by Participants**<br>*Course participants* |
| 13.00 | **Lunch** | **Lunch** | **Lunch** |
| 14.00 | **Use-Case Sprint I**<br>Ideation of Use-Cases<br>Prototype Workflow<br>*Course participants*<br>*Alves Werb, Seltmann DSZ* | **Use-Case Sprint III**<br>Programming and Collaboration<br>*Course participants*<br>*Alves Werb, Seltmann DSZ* | **Short Presentations by Participants and Wrap-Up**<br>*Course participants*<br>*Alves Werb, Seltmann DSZ* |
| 15.00 | | | **Evaluation and Farewell**<br>*Schwarz* |
| 15.30 | City tour and joint dinner | | |
| | **End of day around 8pm (20.00)** | **End of day 3:30pm (15.30)** | **End of seminar 3:30pm (15.30)** |

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# Audience Q&A

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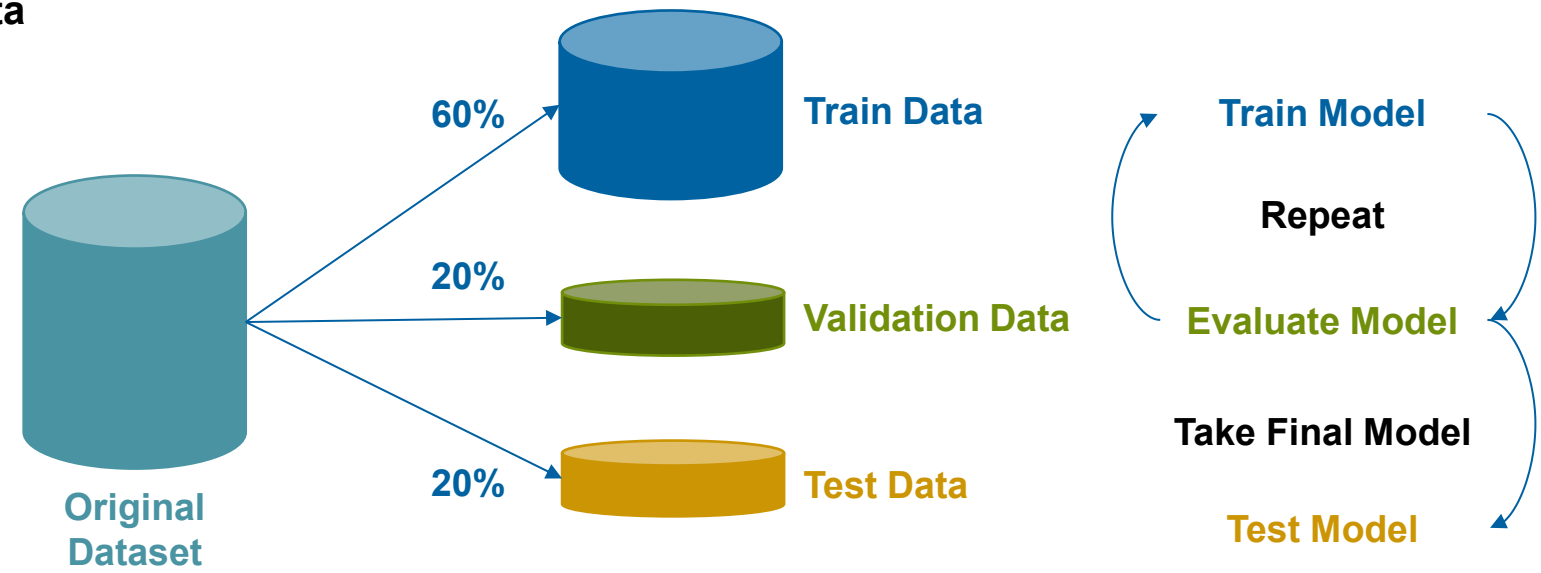
- Ho, J., Jain, A., & Abbeel, P. (2020). *Denoising Diffusion Probabilistic Models*. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2006.11239
- Van den Oord, A., Kalchbrenner, N., & Kavukcuoglu, K. (2016). *Pixel Recurrent Neural Networks*. In *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/1601.06759
- Lipman, Y., Albergo, M., Vanden-Eijnden, E., & Cunningham, W. (2023). *Flow Matching for Generative Modeling*. In *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2209.17123

# Backup

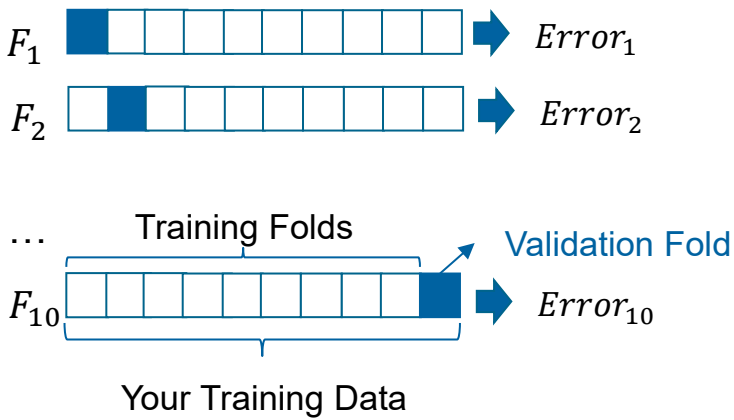
# Train, Validation, and Test Subsamples

- Test data only to be used in the very end
- Performance of the model in this subsample provides an **unbiased estimate** of the **error** on **new data**



# Cross-Validation

- Idea: use **many subsamples of the data** to estimate model error
- K subsamples = K-Fold cross-validation
- Measure error
- Aggregate the errors (e.g., average)

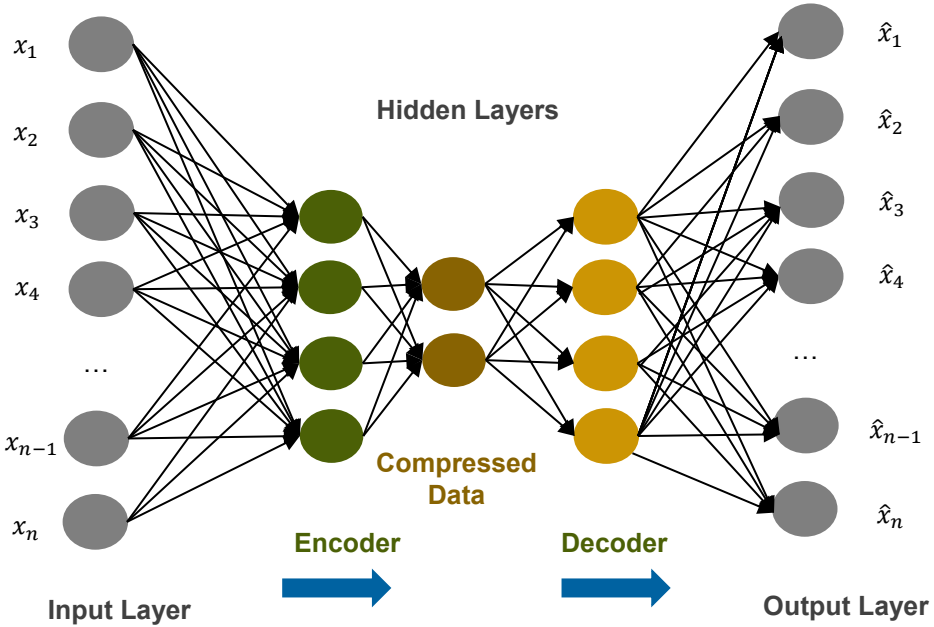


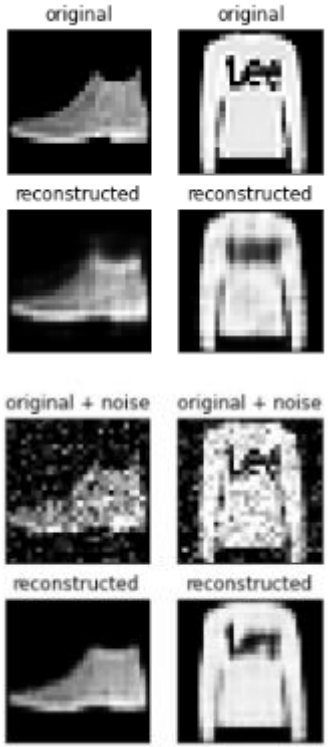

$$\widehat{Error} = \sum_{k=1}^{10} Error_k$$

or

$$\widehat{Error} = \sum_{k=1}^{10} \frac{Error_k}{10}$$

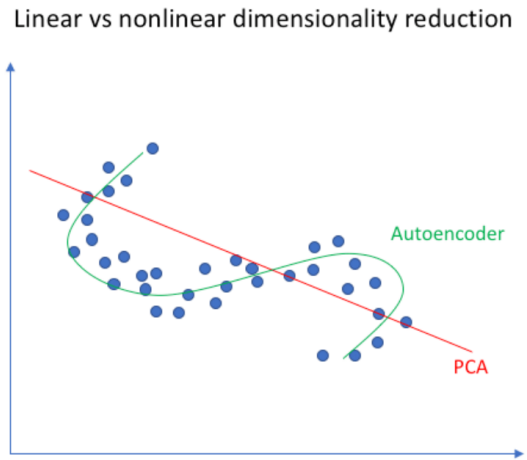
# Autoencoders



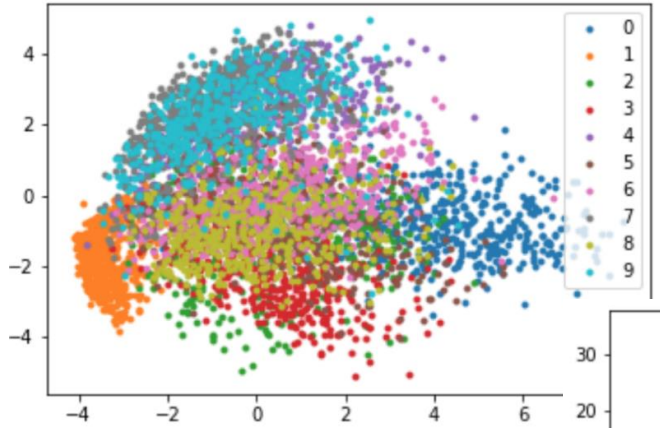




Source: Tensorflow.org

# Autoencoders x Principal Component Analysis (PCA)

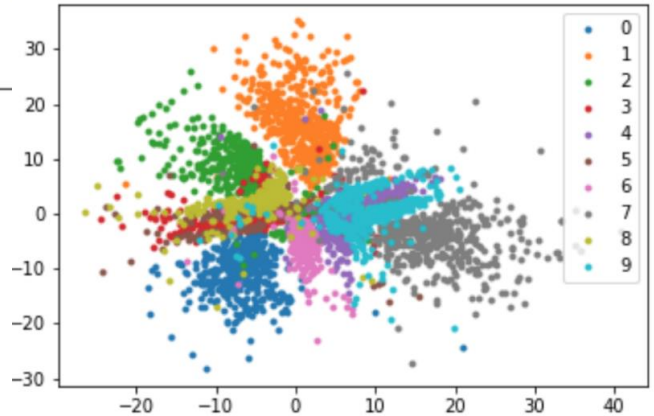



Nugroho (2020)



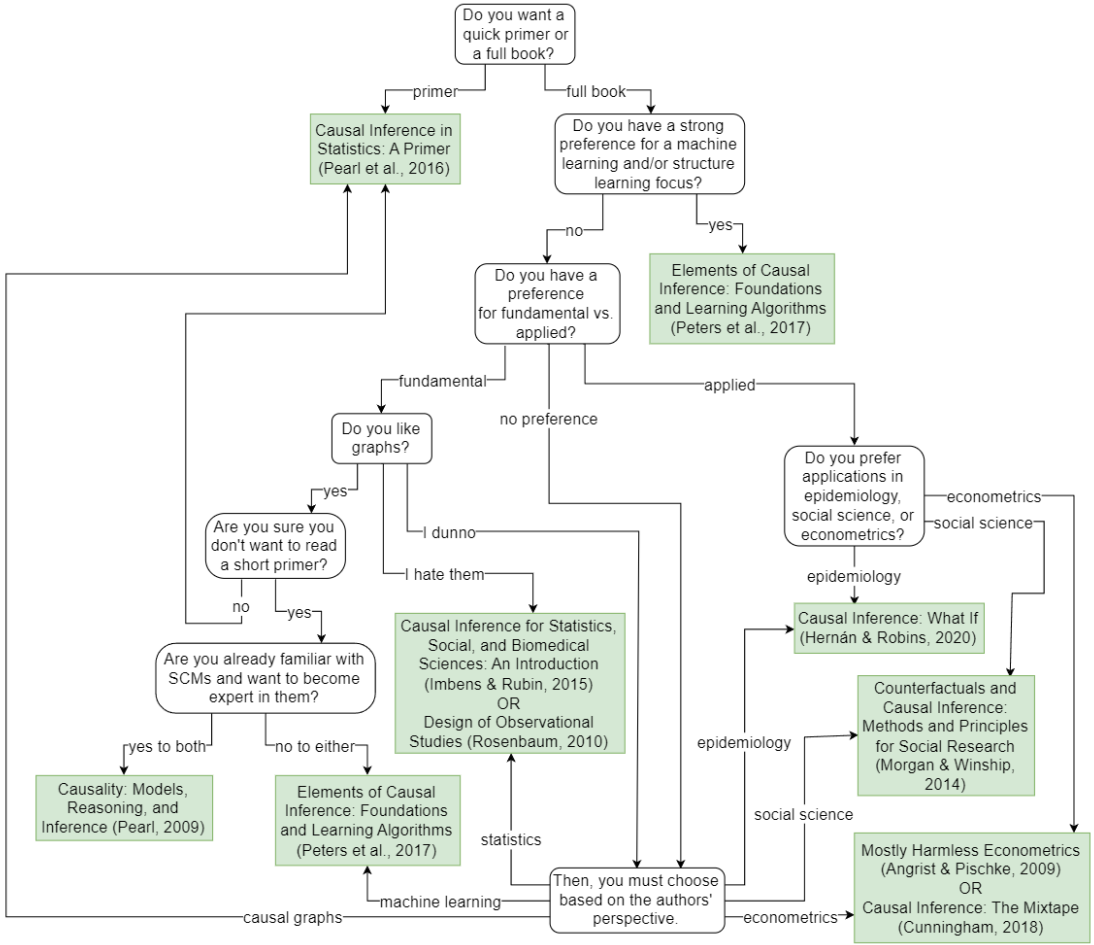

PCA

Tang (2020)




Fine-tuned Autoencoder

# Causal Inference Books

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